Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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Abstract

A d-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ over a commutative semiring $\mathbb{K}$ has *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ equal to the least integer $R \in \mathbb{N}$ for which $K(x) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$ (or ∞ if no finite R exists). On a d-dimensional Cartesian grid of stencil width W, a rank-R kernel evaluates in dRW one-dimensional taps per grid point rather than W^d. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or low-dispersion wave propagators—with finite coordinatewise rank relies on signed profiles built from complex-rate Gaussians and exponential polynomials.

We present a complete machine-checked characterization in Lean 4 (with Mathlib) of the d-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) $\operatorname{srank}_{\mathbb{R}}(K) < \infty$ if and only if the real profile φ agrees on $[0,\infty)$ with a canonical exponential-polynomial representation $\mathcal{R} = \{(\lambda_u, P_u)\}_{u=1}^m$, $\varphi(s) = \sum_{u=1}^m \operatorname{Re}(P_u(s)e^{\lambda_u s})$, with nonzero polynomials $P_u \in \mathbb{C}[X]$, distinct rates λ_u satisfying $\operatorname{Im}\lambda_u \geq 0$, and $P_u \in \mathbb{R}[X]$ whenever $\operatorname{Im}\lambda_u = 0$; (2) over $\mathbb{C}$, a continuous profile ψ on $[0,\infty)$ satisfies $\operatorname{srank}_{\mathbb{C}}(x \mapsto \psi(\|x\|^2)) = R$ if and only if $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ on $[0,\infty)$ for nonzero polynomials $P_\mu \in \mathbb{C}[X]$ at distinct rates $\mu \in S$ with $\sum_{\mu\in S}((d-1)\deg P_\mu + 1) = R$; (3) over $\mathbb{R}$, $\operatorname{srank}_{\mathbb{R}}(x \mapsto \varphi(\|x\|^2)) = C_d(\mathcal{R}) := \sum_{u=1}^m c_d(\lambda_u, P_u)$, where each real-rate component ($\operatorname{Im}\lambda_u = 0$) costs $c_d = (d-1)\deg P_u + 1$ real d-way products (via real Lagrange interpolation) and each nonreal conjugate-pair component ($\operatorname{Im}\lambda_u > 0$) costs $c_d = 2(d-1)\deg P_u + d$ real d-way products (achieved via an alternating root-of-unity polynomial-product identity giving d rather than 2^{d-1} real products for a complex Gaussian when $\deg P_u = 0$, and proved optimal by Alder–Strassen degree peeling over $\mathbb{R}$); and (4) in $d = 2$, scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ preserves rank unconditionally and $\operatorname{srank}_{\mathbb{R}}$ equals the degree of the minimal monic half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$, whereas for every $d \geq 3$ each nonreal conjugate pair incurs a degree-independent real-over-complex rank penalty of $d-2$, and every representation $\mathcal{R}$ is the uniform limit on compact sets of multi-Gaussian kernels of real rank at most $B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$ (strictly below exact rank whenever $\deg P_u \geq 1$).

Necessity rests on the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0,\infty)$: comparing a 96-line Banach-algebra integral-smoothing proof in Lean 4 against a 256-line finite-difference proof and three finite-dimensional nonstandard proofs (206 and 399 lines in Lean 4, 275 lines in Isabelle/HOL) illustrates how Mathlib's Bochner-integral library keeps standard integration shorter and valid in infinite dimensions even after homogeneous scaling $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ eliminates the 124-line difference-quotient bound. Finally, we connect the root-of-unity identity to planar harmonic ("beach-ball") differential operators, evaluate reproducible GPU studies on an NVIDIA RTX 4070 Ti Super (with explicit coefficients in Appendix A) for 2-D circular lens blur ($\mathrm{GL}_R(\mathbb{R})$ orthogonalization for IEEE `float16` sRGB rendering, plus visual and quantitative comparisons against direct discrete rank-R optimization) and 3-D homogeneous-room acoustic leapfrog stencils ($R = 1$–9, identifying Kowalczyk and van Walstijn's 27-point IWB scheme as the $R = 1$ real-Gaussian baseline),

and document practical trade-offs against global `cuFFT` convolution/k-space methods, wide-stencil boundary obstacles, and remaining open questions. All Lean 4 and Isabelle/HOL formalizations, Python/PyTorch/Triton optimization and GPU benchmark scripts, JSON numerical archives, and figure/manuscript build scripts are publicly available under CC0 1.0 at https://github.com/OlliNiemitalo/kernel-rank-lean.

1 Introduction

Convolving a d-dimensional array with a dense $W \times \cdots \times W$ spatial stencil requires W^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional passes, costing dRW operations per sample (or $O(dR)$ operations independent of W when the one-dimensional factors admit recursive causal/anti-causal IIR realizations [19, 76]). Viewing (1) over a commutative semiring $\mathbb{K}$, the *separable rank* $\operatorname{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$ is the least $R \in \mathbb{N}$ for which such a decomposition exists: matrix rank [68] when $d = 2$ and $\mathbb{K}$ is a field, and functional canonical polyadic (CP) or *separation rank* [34, 5, 41, 30] when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

classical characterizations by Maxwell [50] and Kannappan and Sahoo [38] imply that a continuous nonvanishing radial kernel with $\operatorname{srank}_{\mathbb{R}}(K) = 1$ has a Gaussian profile $\varphi(s) = ae^{-\alpha s}$. Real-rate Gaussians ($\alpha > 0$), however, are strictly monotone and convex in the squared radius $s = \|x\|^2$; single Gaussians and nonnegative Gaussian mixtures cannot reproduce flat passbands, sharp cutoffs, or oscillatory diffraction rings [77]. In image processing, real-time graphics, and numerical wave propagation, Niemitalo [54, 57, 58], Garcia [27], and related filter-design studies [45] approximate flat-top and oscillatory radial profiles—such as camera defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\operatorname{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe cancellation while maintaining finite separable rank.

1.1 Visual overview: components, rendering terms, and streaming intermediates

Before introducing technical definitions, Figure 1 illustrates in plain visual terms how one-dimensional filters build a circularly symmetric 2-D kernel and why the distinction between *components* and *rendering terms* governs practical GPU execution:

1. **Mathematical "Components" view (Figure 1a):** Replacing a real decay $-\alpha$ by a complex rate $\lambda = -\alpha + i\omega$ preserves the coordinate split $e^{\lambda(x^2+y^2)} = e^{\lambda x^2} e^{\lambda y^2}$ while oscillating radially as $e^{-\alpha r^2}\cos(\omega r^2)$ and $e^{-\alpha r^2}\sin(\omega r^2)$ ($r^2 = x^2 + y^2$). Writing $g(x) = \sqrt{\beta}\, e^{\lambda x^2} = g_{\mathrm{re}}(x) + i g_{\mathrm{im}}(x)$, taking the real part of $g(x)g(y)$ uses two real 1-D×1-D outer products:

$$\operatorname{Re}\big(g(x)g(y)\big) = g_{\mathrm{re}}(x)\, g_{\mathrm{re}}(y) + g_{\mathrm{im}}(x)\, \big(-g_{\mathrm{im}}(y)\big).$$

   Neither outer product is circular on its own—each is a rectangular plaid of positive (blue) and negative (red) stripes—yet adding the pair cancels the grid corners to form a circularly

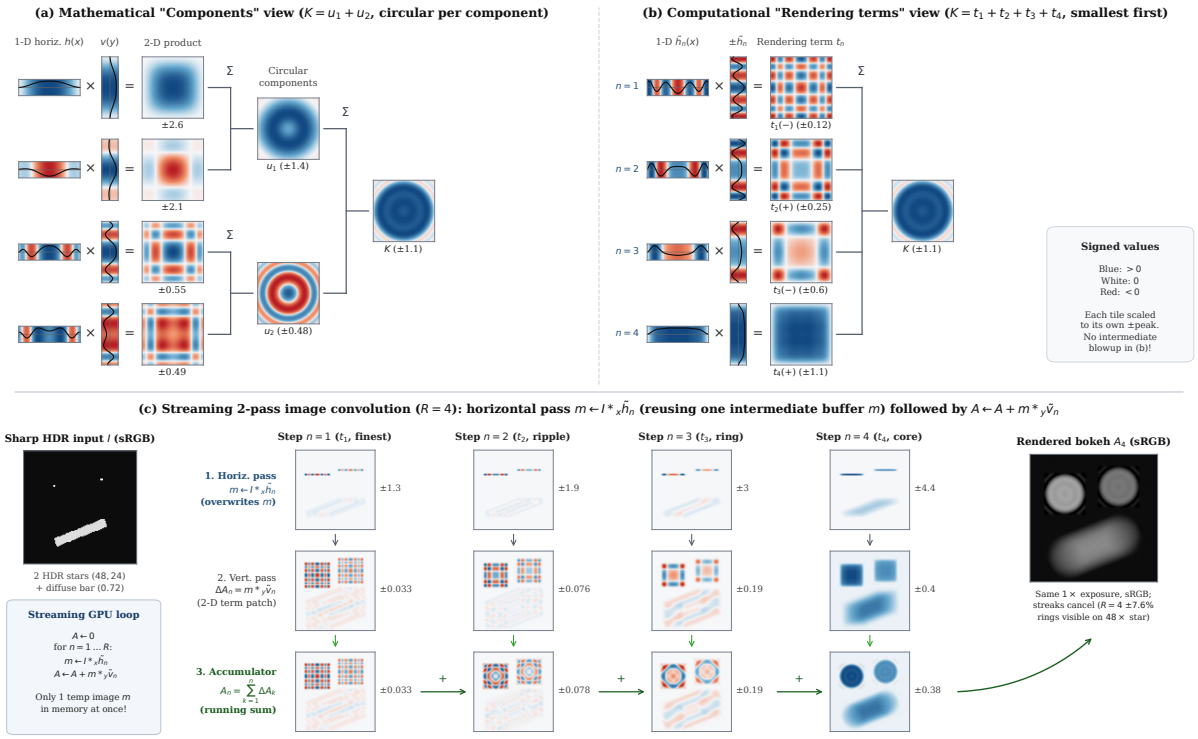


Figure 1: Visual guide to separable radial filtering on a two-complex-Gaussian ($n_{cx} = 2$, exact rank $R = 4$) circular disc-blur kernel [58]: (a) mathematical *Components* view, where pairs of 1-D outer products $h(x)v(y)$ form circularly symmetric concentric-ring components u_1, u_2 that sum to the flat-top disc K; (b) computational *Rendering terms* view, where orthogonalizing rowSpan(K) yields four direct real products $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered smallest-first so intermediate amplitudes stay bounded; (c) streaming two-pass linear-light HDR blur using *one* overwritten horizontal intermediate buffer $m \leftarrow I *_x \tilde{h}_n$ followed by vertical accumulation $A \leftarrow A + m *_y \tilde{v}_n$ (with I and A_4 displayed at identical 1× linear exposure through the IEC 61966-2-1 sRGB transfer function; all axis-aligned streaks cancel, while $R = 4$'s ±7.6% pass/stopband ripple remains visible on the 48× HDR star).

symmetric component (u_1: wide dome; u_2: concentric sharpening rings), whose sum $u_1 + u_2$ approximates a flat-top disc K. (When a component multiplies $e^{\lambda r^2}$ by a polynomial $P(r^2)$ of degree k, expanding $(x^2 + y^2)^q$ naively across monomials would produce $(k + 1)(k + 2)$ real products, whereas collecting powers of x^2 merges them into $2(k + 1)$ real products, or $k + 1$ when $\omega = 0$.)

2. **Computational "Rendering terms" view (Figure 1b):** A renderer never computes the circular components u_1, u_2 separately—only the total sum $K(x, y) = \sum_{n=1}^{R} \tilde{h}_n(x)\tilde{v}_n(y)$. In high-accuracy approximations (such as $n_{cx} = 6$ complex Gaussians, $R = 12$), individual components u_i grow hundreds of times taller than K and cancel against each other, causing severe precision loss in 16-bit half-float (`float16`) GPU buffers. Orthogonalizing the 1-D filter subspace (`separableChannels` in [58]; Section 6.3) rotates the R filter pairs into orthogonal *rendering terms* $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered from smallest to largest magnitude ($n = 1, \ldots, 4$): every term stays comparable to the kernel's own peak (±0.12 to ±1.1 in Figure 1b, vs. ±2.6 in Figure 1a).

3. **Streaming two-pass convolution with one intermediate buffer (Figure 1c):** To blur a sharp input image $I(x, y)$, the GPU initializes an output accumulator $A \leftarrow 0$ and loops over $n = 1, \ldots, R$ (smallest term first). In step n, a 1-D horizontal pass writes

Table 1: Overview of main theoretical, formalization, and numerical results ($d \geq 2$, degree $k = \deg P$).

Regime	Setting	Main result	Ref.
Finite-rank form	Cont. φ on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} < \infty \iff \varphi(s) = \sum_u \operatorname{Re}(P_u(s)e^{\lambda_u s})$ $(s \geq 0)$	Thm. 2.3
Complex rank	$\psi : \mathbb{R} \to \mathbb{C}$, $\mathbb{R}^d$	$(d-1)k+1$ per distinct rate $\mu \in S$	Thm. 2.4
Real rank (Im λ=0)	$P \in \mathbb{R}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = (d-1)k+1$ $(= 1$ at $k = 0)$	Thm. 2.5
Real rank (Im λ>0)	$P \in \mathbb{C}[X]$, $\mathbb{R}^d$	$c_d(\lambda, P) = 2(d-1)k+d$ $(= d$ vs. 2^{d-1} at $k = 0)$	Thm. 2.5, Prop. 2.9
Real–complex gap	Conjugate pair	$c_d(\lambda, P) - 2((d-1)k+1) = d-2$ (all $k \geq 0$)	Rem. 2.11
Border-rank bound	Any list $\mathcal{R}$	Border rank $\leq B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$	Thm. 2.7
2-D diff. order	$d = 2$, D on $\mathbb{R}_{\geq 0}$	$\operatorname{srank}_{\mathbb{R}} = \deg p_{\min}$ for minimal $p_{\min}(D)\varphi_{\mathbb{C}} = 0$	Thm. 2.6
Semigroup proofs	$M(t) = \exp(tA)$	Integral (96 loc, Banach) vs. 4 FD/NSA (206–399 loc)	Secs. 4–5
GPU studies	Blur; 3-D wave	`fp16` sRGB $\leq 0.33/255$; $\lvert\delta c/c_0\rvert \leq 0.049\%$, vs. `cuFFT`	Secs. 6–7, App. A

$m(x, y) \leftarrow (I *_x \tilde{h}_n)(x, y)$, smearing bright point stars horizontally into 1-D signed streaks; crucially, *the same temporary buffer m is overwritten on every step*, keeping intermediate memory constant at one image regardless of R. A 1-D vertical pass then adds $\Delta A_n = m *_y \tilde{v}_n$ into A: steps $n = 1, 2, 3$ build a hollow sharpening ring A_3, and step $n = 4$ adds the soft core so all horizontal and vertical streaks cancel in A_4.

1.2 Plain-language summary of main results and paper organization

Table 1 summarizes our results, organized into three parts so readers can navigate Mathematical Theory (Sections 2–3), Formal Verification (Sections 4–5), and Applications & Limitations (Sections 6–7, Appendix A) independently:

1. **Part I — Mathematical characterization and exact ranks (Sections 2–3):** First, exponential-polynomial profiles in $s = \|x\|^2$ are *universal*: no other continuous radial profile on $\mathbb{R}^d$ $(d \geq 2)$ admits finite coordinatewise separable rank (Theorem 2.3). Second, we establish closed-form exact d-way separable ranks over $\mathbb{C}$ (Theorem 2.4) and over $\mathbb{R}$ (Theorem 2.5): in $d \geq 3$, an alternating root-of-unity identity reduces each complex Gaussian from 2^{d-1} naive real products to d (3 instead of 4 in 3-D) and each degree-k nonreal component to $2(d-1)k+d$ real products, incurring a constant $d-2$ real-over-complex penalty independent of k. Third, we prove a general multi-Gaussian approximation theorem (Theorem 2.7) showing that every radial profile with $\deg P_u \geq 1$ in $d \geq 3$ has compact-open real border rank strictly smaller than its exact real rank by at least $(d-2)\sum_u \deg P_u$.

2. **Part II — Semigroup formalizations and proof engineering (Sections 4–5):** All mathematical theorems in Sections 2–4 (22 top-level Lean 4 theorems and 1 Isabelle/HOL theorem) are machine-checked without unproved steps. Section 4 compares five formalizations of the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$, introducing a projective normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ that removes the 124-line difference-quotient bound from Euler's infinitesimal proof while showing why Mathlib's library-amortized Bochner integration remains 2×–4× shorter and applies to infinite-dimensional Banach algebras. Section 5 documents the module architecture, theorem-to-declaration map, `noncomputable` infimum semantics, and classical axioms.

3. **Part III — Analytical/GPU applications, limitations, and reproducible tables (Sections 6–7, Appendix A):** Section 6 unifies odd and even planar "beach-ball" differential operators, proves two-stage steerable pass-count lower bounds, and evaluates GPU-verified 2-D lens blur and 3-D acoustic wave propagation on an NVIDIA RTX 4070 Ti Super. Section 7 examines continuous-domain scope, root-of-unity numerical conditioning, and open border-rank questions, while Appendix A tabulates explicit filter and propagator coefficients for standalone reproduction from this PDF.

1.3 Related work

Classical radial characterizations, continuous Kronecker theorems, and functional equations. Maxwell [50] and Kannappan and Sahoo [38] proved that measurable or continuous rotation-invariant rank-1 functions are Gaussians, while Schoenberg [63] characterized completely monotone radial profiles as positive Gaussian mixtures. Under the coordinate surjection $s_j = x_j^2$, radial separability on $\mathbb{R}^d$ becomes the d-variable additive Hankel functional equation $\varphi(s_0 + \cdots + s_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} g_{n,j}(s_j)$ on $[0,\infty)^d$. In two variables, functional equations $f(x+y) = \sum_{n=1}^{R} X_n(x)Y_n(y)$ go back to Levi-Civita [47], Stone [64], Anselone and Korevaar's characterization of finite-dimensional translation-invariant subspaces [3], and Laczkovich [44]. In operator theory, Rochberg [62] and Andersson and Carlsson [2] established continuous-domain analogues of Kronecker's 1881 theorem for finite-rank truncated Hankel and convolution integral operators; whereas bounded L^2 integral-operator formulations assume square-integrability or bounded domains, our pointwise algebraic formulation on $[0,\infty)^d$ characterizes all continuous profiles on $[0,\infty)$—including undamped or polynomially growing profiles such as $\|x\|^2\cos(\|x\|^2)$—via operator-norm continuous shift semigroups without assuming differentiability or decay.

Separation rank, RBF kernels, and exponential-sum approximations. Beylkin and Mohlenkamp [5], Hackbusch [30], and Khoromskij [39] develop tensor-product and separation-rank representations for multivariate radial operators, building on nonlinear exponential-sum approximation theory [12, 29, 13, 6, 48] and skeleton/cross approximations of smooth radial kernels [69, 73, 67]. Two distinctions clarify how our results complement this literature: (i) a pairwise RBF kernel $(x,y) \in \mathbb{R}^d \times \mathbb{R}^d \mapsto e^{-\alpha\|x-y\|^2}$ has *infinite* exact two-argument rank on $\mathbb{R}^d \times \mathbb{R}^d$ (since $e^{2\alpha x\cdot y}$ is not a finite sum of products $u_n(x)v_n(y)$), whereas as a *convolution stencil* $z \mapsto \varphi(\|z\|^2)$ in $z = x - y$ its coordinatewise d-way rank across $(z_0,\dots,z_{d-1})$ is 1; and (ii) prior sum-of-Gaussians methods construct upper-bound approximations, whereas Theorems 2.3–2.5 establish the exact converse and exact d-way real CP rank for all signed/oscillatory exponential-polynomial profiles.

Algebra multiplicative complexity, Hankel/Waring tensors, real vs. complex gaps, and border rank. Once necessity identifies the $\mathbb{K}[D]$-module $\mathcal{W}(f)$ with the commutative quotient algebra $\mathcal{A} = \mathbb{K}[X]/(p_{\min,\mathbb{K}}(X))$ (where $p_{\min,\mathbb{K}}$ has m_{irr} distinct irreducible factors over $\mathbb{K}$), the multilinear core of our d-way rank problem is the $(d-1)$-fold multiplication tensor $\mathcal{A}^{d-1} \times \mathcal{A}^* \to \mathbb{K}$, $(u_1,\dots,u_{d-1},\ell) \mapsto \ell(u_1\cdots u_{d-1})$. At $d = 3$ (2-fold bilinear multiplication), Winograd [74], Fiduccia and Zalcstein [23], and Alder and Strassen [1] (with nilradical extensions by Bläser [9]; see Bürgisser, Clausen, and Shokrollahi [15], Thm. 17.20) proved that multiplying in $\mathbb{K}[X]/(p)$ requires $2\dim_{\mathbb{K}}\mathcal{A} - m_{\mathrm{irr}}$ bilinear multiplications, yielding our $d = 3$ costs $2k+1$ and $4k+3$. Recently, Bläser, Mayer, and Shringi [10] generalized the Alder–Strassen lower bound to $(d-1)$-fold multiplication in arbitrary associative algebras, establishing the lower bound $(d-1)\dim_{\mathbb{K}}\mathcal{A} - (d-2)m_{\mathrm{irr}}$. Theorems 2.4 and 2.5 connect continuous radial kernels on $\mathbb{R}^d$ to this invariant, prove that $\mathbb{K}[X]/(p_{\min,\mathbb{K}})$ attains the Bläser–Mayer–Shringi bound *with equality* over both $\mathbb{C}$ and $\mathbb{R}$ for all $d \geq 2$ via explicit root-of-unity functional syntheses (Propositions 2.8–2.9, without solving Vandermonde systems), and machine-check both the lower and upper bounds in Lean 4. This complements apolarity and discrete Hankel/Waring tensor theory [17, 36, 43, 53], real-versus-complex rank separations [18, 20, 61, 11], border-rank gaps on W-type and local-algebra tensors [8, 46, 14, 78], and border-subrank results on higher-order algebra structure tensors [16].

Steerable filters, acoustic FDTD stencils, semigroups, and formalized mathematics. In vision and graphics, multi-stage separable planar filters [68], steerable and complex-derivative

filters [26, 40, 59, 7, 60, 71], anisotropic Gaussians [45], and complex-Gaussian circular bokeh filters [54, 27, 55, 56, 57, 58] motivate our analytical and GPU blur studies; in computational acoustics and geophysics, dispersion-relation-preserving and k-space methods [66, 65], compact 27-point and higher-order 3-D room-acoustics FDTD stencils [42, 72, 31, 32], and low-rank wave extrapolators [25] provide our 3-D wave-propagation baselines. Finally, our semigroup comparison builds on classical Banach-algebra semigroups [52, 22], nonstandard Lie theory [33, 28], Lean 4/Mathlib [51, 49, 21], Isabelle/HOL analysis and NSA [24, 35, 70], and recent formalizations of tensor rank [4, 37, 75].

2 Problem Formulation, Harmonized Notation, and Main Theorems

2.1 Separable rank, coordinate surjections, and canonical representations

Throughout the paper, we adhere to one uniform profile convention: $\varphi : \mathbb{R} \to \mathbb{R}$ denotes a *real-valued* radial profile, $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$ denotes its complexification, $\psi : \mathbb{R} \to \mathbb{C}$ denotes a general *complex-valued* radial profile, and $f : \mathbb{R} \to \mathbb{K}$ denotes a profile taking values in an arbitrary commutative semiring or field $\mathbb{K}$. Similarly, for any kernel $K : X \to \mathbb{R}$, we write $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ for its scalar extension to $\mathbb{C}$.

Definition 2.1 (Coordinatewise separable rank; Lean: `IsSumOfProducts`, `separableRank`)**.** Let ι be a finite index set, $(X_i)_{i\in\iota}$ a family of sets, and $\mathbb{K}$ a commutative semiring. A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of R products* ($R \in \mathbb{N}$) if there exist univariate factors $f_{n,i} : X_i \to \mathbb{K}$ ($0 \le n < R$, $i \in \iota$) such that $K(x) = \sum_{n=0}^{R-1} \prod_{i\in\iota} f_{n,i}(x_i)$ for all $x \in \prod_{i\in\iota} X_i$ (taking the empty sum 0 when $R = 0$). Its *separable rank* over $\mathbb{K}$ is the extended natural number

$$\operatorname{srank}_{\mathbb{K}}(K) := \inf\big\{R \in \mathbb{N} : K \text{ is a sum of } R \text{ products over } \mathbb{K}\big\} \in \mathbb{N} \cup \{\infty\},$$

where $\inf \emptyset = \infty$, and an equation $\operatorname{srank}_{\mathbb{K}}(K) = R$ with $R \in \mathbb{N}$ denotes equality in $\mathbb{N} \cup \{\infty\}$ under the canonical inclusion $\mathbb{N} \hookrightarrow \mathbb{N} \cup \{\infty\}$.

Separable rank is subadditive under finite sums and cannot increase when we substitute coordinate maps $y_i \mapsto g_i(y_i)$ into an existing decomposition $\sum_n \prod_i f_{n,i}(g_i(y_i))$. When every $g_i : Y_i \to X_i$ ($i \in \iota$) is *onto* (surjective), the reverse inequality holds as well: picking for each $x_i \in X_i$ at least one preimage $h_i(x_i) \in Y_i$ with $g_i(h_i(x_i)) = x_i$ turns any decomposition $\sum_n \prod_i \tilde{f}_{n,i}(y_i)$ of $K(g_0(y_0), \dots)$ back into $\sum_n \prod_i \tilde{f}_{n,i}(h_i(x_i)) = K(x)$ with the same number of terms:

$$\operatorname{srank}_{\mathbb{K}}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \operatorname{srank}_{\mathbb{K}}(K). \tag{2}$$

For $x_j \in \mathbb{R}$, the squaring map $x_j \mapsto x_j^2$ covers $\mathbb{R}_{\ge 0} = [0, \infty)$ (with right inverse $s_j \mapsto \sqrt{s_j}$). Applying (2) therefore equates the d-way separable rank of any radial kernel on $\mathbb{R}^d$ with that of the d-way additive Hankel kernel on $\mathbb{R}^d_{\ge 0}$:

$$\operatorname{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \operatorname{srank}_{\mathbb{K}}\left(s \in \mathbb{R}^d_{\ge 0} \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

Notice two immediate consequences of (3): (i) because $\|x\|^2 \ge 0$ never visits negative arguments, $K(x) = \varphi(\|x\|^2)$ depends only on $\varphi|_{[0,\infty)}$ and leaves $\varphi(s)$ for $s < 0$ completely unconstrained; and (ii) replacing $x_j \mapsto x_j^2$ by $x_j \mapsto x_j^2/a_j^2$ ($a_j \ne 0$) or $x_j \mapsto |x_j|^p$ ($p > 0$) is still onto $[0, \infty)$, so every theorem below holds verbatim for axis-aligned ellipsoidal kernels $f(\sum_j x_j^2/a_j^2)$ and ℓ_p superellipsoidal kernels $f(\sum_j |x_j|^p)$. Finally, for $d \ge 2$, setting $x_2 = \cdots = x_{d-1} = 0$ turns

Table 2: Harmonized symbol glossary and cost specializations ($d \geq 2$, $k = \deg P$).

Mathematical symbol	Lean 4 identifier	Meaning / specialization
$\mathrm{srank}_{\mathbb{K}}(K) \in \mathbb{N} \cup \{\infty\}$	`separableRank K (ENat)`	Coordinatewise separable rank over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$
φ, $\varphi_{\mathbb{C}}$, ψ, f	$\varphi : \mathbb{R} \to \mathbb{R}$, $\psi : \mathbb{R} \to \mathbb{C}$	Real profile, complexification, complex profile, $\mathbb{K}$-profile
$W = 2L + 1$; $s = \|x\|^2$	—; $\sum_j$ `x j ^ 2`	1-D discrete tap width on $[-L, L]$; squared radius on $\mathbb{R}^d$
$\mathcal{W}(f)$	`translateSpan f`	Half-line translate span $\mathrm{span}_{\mathbb{K}}\{u \mapsto f(u+t) : t \geq 0\}$
(λ_u, P_u); $\lambda_u = -\alpha_u + i\omega_u$	`Component.mk rate poly`	Well-formed: $P_u \neq 0$, $\omega_u \geq 0$, $\omega_u{=}0 \Rightarrow P_u \in \mathbb{R}[X]$
n_{re}, n_{cx}; m_{irr}	—	Real and nonreal component counts; $m_{\mathrm{irr}} = n_{\mathrm{re}} + n_{\mathrm{cx}}$
$C_d(\mathcal{R}) = \sum_u c_d(\lambda_u, P_u)$	`multiTotalCost d cs`	Exact d-D real radial rank (Theorem 2.5)
$C_2(\mathcal{R})$; $n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$	`totalCost`; `gaussianTotalCost`	Specializations at $d = 2$ and at $\deg P_u = 0$
$B_d(\mathcal{R}) = \sum_u b_d(\lambda_u, P_u)$	`borderTotalCost d cs`	Multi-Gaussian border bound $C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$
$H_P(s_0; X)$; $E_k(s_j; X)$	`revHassePoly`; `truncExpPoly`	Reversed Hasse poly.; truncated exp $\sum_{q=0}^{k} s_j^q X^q/q!$
$M_{d,k}^{\mathbb{C}}$, $M_{d,k}^{\mathbb{R}}$	$(d-1)k+1$, $2(d-1)k+d$	Complex and nonreal root-of-unity node counts

each d-way product $\prod_{j=0}^{d-1} f_{n,j}(x_j)$ into the 2-way product $(c_n f_{n,0}(x_0)) f_{n,1}(x_1)$ with constant $c_n = \prod_{j=2}^{d-1} f_{n,j}(0)$, giving the 2-D slice lower bound:

$$\mathrm{srank}_{\mathbb{K}}\big(y \in \mathbb{R}^2 \mapsto f(y_0^2 + y_1^2)\big) \leq \mathrm{srank}_{\mathbb{K}}\left(x \in \mathbb{R}^d \mapsto f\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

Definition 2.2 (Canonical real representations and cost functions)**.** A *component* (`Component`) is a pair $(\lambda, P) \in \mathbb{C} \times \mathbb{C}[X]$, evaluating at $s \in \mathbb{R}$ as $\mathrm{Re}(P(s)e^{\lambda s})$ of degree $k = \deg P$ (multiplicity $k+1$). It is *well-formed* if $P \neq 0$, $\mathrm{Im}\,\lambda \geq 0$, and $\mathrm{Im}\,\lambda = 0 \implies P \in \mathbb{R}[X]$. A finite list $\mathcal{R} = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ is a *representation* (`IsRepresentation`) if every (λ_u, P_u) is well-formed and the rates $\lambda_1, \ldots, \lambda_m$ are pairwise distinct; its radial profile is $\mathrm{profile}(\mathcal{R}, s) = \sum_{u=1}^{m} \mathrm{Re}(P_u(s)e^{\lambda_u s})$. For $d \geq 2$, define the exact d-way component cost $c_d(\lambda, P)$ and multi-Gaussian border-rank cost $b_d(\lambda, P)$ by

$$c_d(\lambda, P) = \begin{cases} (d-1)\deg P + 1, & \mathrm{Im}\,\lambda = 0, \\ 2(d-1)\deg P + d, & \mathrm{Im}\,\lambda > 0, \end{cases} \qquad b_d(\lambda, P) = \begin{cases} \deg P + 1, & \mathrm{Im}\,\lambda = 0, \\ d(\deg P + 1), & \mathrm{Im}\,\lambda > 0, \end{cases} \tag{5}$$

with total costs $C_d(\mathcal{R}) = \sum_{u=1}^{m} c_d(\lambda_u, P_u)$ (`multiTotalCost`) and $B_d(\mathcal{R}) = \sum_{u=1}^{m} b_d(\lambda_u, P_u)$ (`borderTotalCost`). When every $\deg P_u = 0$ (*multi-Gaussian* representation with n_{re} real and n_{cx} nonreal rates), $C_d(\mathcal{R}) = B_d(\mathcal{R}) = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$.

Why are the conditions $\mathrm{Im}\,\lambda \geq 0$ and ($\mathrm{Im}\,\lambda = 0 \implies P \in \mathbb{R}[X]$) canonical? Because $\mathrm{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \mathrm{Re}(P(s)e^{\lambda s})$ and $\mathrm{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs over-parameterize real profiles. Under Definition 2.2, each real rate ($\mathrm{Im}\,\lambda = 0$) contributes a single term $P(s)e^{\lambda s}$ over $\mathbb{C}$, while each upper-half-plane rate ($\mathrm{Im}\,\lambda > 0$) expands via $\mathrm{Re}\, z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$. Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$, every continuous real profile of finite rank determines its unordered set of well-formed components uniquely. Table 2 summarizes all symbols and their Lean 4 counterparts.

2.2 Statement of main theorems and algebraic identities

Theorem 2.3 (d-D Finite-Rank Classification; Lean: `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. Then* $\mathrm{srank}_{\mathbb{R}}\big(x \in \mathbb{R}^d \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)\big) < \infty$ *if and only if there exists a representation $\mathcal{R}$ such that $\varphi(s) = \mathrm{profile}(\mathcal{R}, s)$ for all $s \geq 0$.*

Theorem 2.4 (d-D Complex Radial Rank; Lean: `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{C}} \left(x \in \mathbb{R}^d \mapsto \psi \left(\sum_{j=0}^{d-1} x_j^2 \right) \right) = R$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and nonzero polynomials $(P_\mu)_{\mu \in S} \subset \mathbb{C}[X]$ such that $\psi(s) = \sum_{\mu \in S} P_\mu(s) e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu \in S} \big((d-1) \deg P_\mu + 1 \big) = R$.

Theorem 2.5 (Exact d-D Real Radial Rank; Lean: `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $R \in \mathbb{N}$,*

$$\operatorname{srank}_{\mathbb{R}} \left(x \in \mathbb{R}^d \mapsto \varphi \left(\sum_{j=0}^{d-1} x_j^2 \right) \right) = R$$

if and only if there exists a representation $\mathcal{R}$ such that $\varphi(s) = \operatorname{profile}(\mathcal{R}, s)$ for all $s \geq 0$ and $C_d(\mathcal{R}) = R$. In particular, for a multi-Gaussian representation with n_{re} real and n_{cx} nonreal components, $\operatorname{srank}_{\mathbb{R}} = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$, and for $d = 2$, $\operatorname{srank}_{\mathbb{R}} = C_2(\mathcal{R}) = \operatorname{srank}_{\mathbb{C}}(x \mapsto \varphi_{\mathbb{C}}(x_0^2 + x_1^2))$.

Theorem 2.6 (2-D Minimal Differential Annihilator; Lean: `exists_minimal_differential_annihilator`)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with $\operatorname{srank}_{\mathbb{R}}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = R < \infty$, and let D denote differentiation within $[0, \infty)$ (right differentiation at 0). There exists a unique monic polynomial $p_{\min} \in \mathbb{C}[X]$ of degree R such that for every $q \in \mathbb{C}[X]$, $q(D)\varphi_{\mathbb{C}} = 0$ on $[0, \infty)$ if and only if $p_{\min} \mid q$.*

In $d = 2$ (matrices), a limit of rank-B matrices always has rank at most B. In $d \geq 3$ dimensions, however, a kernel of exact rank R can be the limit of kernels of strictly smaller rank $B < R$ whose weights grow as $O(1/h)$ and cancel as $h \to 0$—just as $s_0 + s_1 + s_2 = \lim_{h \to 0} h^{-1}\big((1 + hs_0)(1 + hs_1)(1 + hs_2) - 1\big)$ approximates a rank-3 sum by a difference of *two* 3-way products [8, 46]. The least $B \in \mathbb{N}$ for which K is a uniform limit on every compact set $C \subset \mathbb{R}^d$ of kernels of rank $\leq B$ is the *compact-open border rank* of K:

Theorem 2.7 (General d-D Multi-Gaussian Border-Rank Bound and Gap)**.** *Let $d \geq 2$ and let $\mathcal{R} = [(\lambda_1, P_1), \ldots, (\lambda_m, P_m)]$ be any finite list of well-formed components. Formalized as `multiTotalCost_eq_borderTotalCost_add` and `exists_rank_le_borderTotalCost_tendstoUniformlyOn_radial_profile`:*

1. Exact gap identity: *Both real and nonreal components satisfy $c_d(\lambda_u, P_u) = b_d(\lambda_u, P_u) + (d-2) \deg P_u$, giving*

$$C_d(\mathcal{R}) = B_d(\mathcal{R}) + (d-2) \sum_{u=1}^{m} \deg P_u. \tag{6}$$

2. Uniform multi-Gaussian approximation on compact sets: *There exists an explicit family of multi-Gaussian kernels $F_h : \mathbb{R}^d \to \mathbb{R}$ ($h \in \mathbb{R} \setminus \{0\}$, extended to $h = 0$ in Lean via $0^{-1} = 0$) satisfying $\operatorname{srank}_{\mathbb{R}}(F_h) \leq B_d(\mathcal{R})$ for all h and $F_h(x) \to \operatorname{profile}(\mathcal{R}, \|x\|^2)$ uniformly on every compact subset $C \subset \mathbb{R}^d$ as $h \to 0$ ($h \neq 0$). Consequently, whenever $\mathcal{R}$ is a representation in dimension $d \geq 3$ with $\sum_u \deg P_u \geq 1$, its compact-open real border rank is strictly less than its exact real rank $C_d(\mathcal{R})$ by at least $(d-2) \sum_u \deg P_u \geq d - 2$. In particular, for $K_d(x) = \|x\|^2 \cos(\|x\|^2)$ ($\mathcal{R} = [(i, X)]$), $\operatorname{srank}_{\mathbb{C}}((K_d)_{\mathbb{C}}) = 2d$, $\operatorname{srank}_{\mathbb{R}}(K_d) = 3d - 2$, and $\operatorname{srank}_{\mathbb{R}}(F_h) \leq 2d$.*

The upper bounds for nonreal components in Theorems 2.5 and 2.7 rest on two root-of-unity identities of independent algebraic interest:

Proposition 2.8 (Scalar alternating $2d$-th root-of-unity identity; Lean: `re_prod_eq_sum_re`)**.** *For every integer $d \geq 1$ and all $z_0, \ldots, z_{d-1} \in \mathbb{C}$,*

$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{r=0}^{d-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(z_j e^{-i\pi r/d}\right). \tag{7}$$

Proposition 2.9 (Universal polynomial-product root-of-unity identity; Lean: `re_coeff_prod_eq_sum_re`)**.** *Let $d \geq 2$, $k \geq 0$, and $M^{\mathbb{R}}_{d,k} = 2(d-1)k + d$. For any polynomials $U_0, \ldots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \leq k$,*

$$\operatorname{Re}\left([X^k] \prod_{j=0}^{d-1} U_j(X)\right) = \frac{2^{d-1}}{M^{\mathbb{R}}_{d,k}} \sum_{r=0}^{M^{\mathbb{R}}_{d,k}-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\left(U_j\left(e^{\frac{2\pi i r}{M^{\mathbb{R}}_{d,k}}}\right) e^{-\frac{i\pi r(2k+1)}{M^{\mathbb{R}}_{d,k}}}\right). \tag{8}$$

Example 2.10 (Concrete $d = 3$ complex Gaussian: 3 real products vs. 4 naive products)**.** Let $d = 3$ and write $z_j = a_j + ib_j \in \mathbb{C}$ ($a_j, b_j \in \mathbb{R}$) for $z_j(x_j) = \beta^{1/3} e^{(-\alpha+i\omega)x_j^2}$. Expanding $\operatorname{Re}(z_0 z_1 z_2)$ directly into real and imaginary parts requires $2^{3-1} = 4$ real 3-way products:

$$\operatorname{Re}(z_0 z_1 z_2) = a_0 a_1 a_2 - a_0 b_1 b_2 - b_0 a_1 b_2 - b_0 b_1 a_2.$$

Instead, Proposition 2.8 at $d = 3$ evaluates $u_{j,r} = \operatorname{Re}(z_j e^{-i\pi r/3}) = a_j \cos(\frac{\pi r}{3}) + b_j \sin(\frac{\pi r}{3})$ at the three angles $0, \frac{\pi}{3}, \frac{2\pi}{3}$ ($u_{j,0} = a_j$, $u_{j,1} = \frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j$, $u_{j,2} = -\frac{1}{2}a_j + \frac{\sqrt{3}}{2}b_j$):

$$\operatorname{Re}(z_0 z_1 z_2) = \frac{4}{3}\Big(u_{0,0}u_{1,0}u_{2,0} - u_{0,1}u_{1,1}u_{2,1} + u_{0,2}u_{1,2}u_{2,2}\Big).$$

Expanding the right-hand side confirms that all $a_0 a_1 b_2$ and $b_0 b_1 b_2$ terms cancel via $-\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{8} = 0$ and $-\frac{3\sqrt{3}}{8} + \frac{3\sqrt{3}}{8} = 0$, while $a_0 a_1 a_2$ keeps $1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}$, leaving $\frac{4}{3}(\frac{3}{4}a_0 a_1 a_2 - \frac{3}{4}(a_0 b_1 b_2 + b_0 a_1 b_2 + b_0 b_1 a_2)) = \operatorname{Re}(z_0 z_1 z_2)$ using *three* symmetric real 3-way products rather than four, and Theorem 2.5 proves that two never suffice when $\omega > 0$. Algebraically, pairing $(z_0, z_1) \mapsto z_0 z_1$ against $\bar{z}_2 \mapsto \operatorname{Re}((z_0 z_1) z_2)$ identifies this $2 \times 2 \times 2$ real tensor with bilinear multiplication in $\mathbb{C} \cong \mathbb{R}[X]/(X^2 + 1)$, whose real rank is 3 by Gauss/Karatsuba complex multiplication [74].

Remark 2.11 (Unified commutative-algebra invariant and comparison with symmetric Waring rank)**.** Write $\mathcal{A}_\varphi = \mathbb{R}[X]/(p_{\min,\mathbb{R}}(X))$ for the finite-dimensional commutative polynomial quotient algebra associated to the minimal real differential annihilator $p_{\min,\mathbb{R}}(D)\varphi = 0$ (isomorphic to $\mathcal{W}(\varphi)$ as an $\mathbb{R}[D]$-module, with multiplication induced by composition of D rather than pointwise product of profiles). Over $\mathbb{R}$, $p_{\min,\mathbb{R}}(X)$ factors into $m_{\text{irr}} = n_{\text{re}} + n_{\text{cx}}$ distinct powers of irreducible real polynomials (n_{re} linear factors $(X - \lambda_u)^{k_u+1}$ and n_{cx} irreducible real quadratics $((X + \alpha_u)^2 + \omega_u^2)^{k_u+1}$; equivalently, m_{irr} maximal ideals of $\mathcal{A}_\varphi$), while $\dim_{\mathbb{R}} \mathcal{A}_\varphi = \deg p_{\min,\mathbb{R}} = C_2(\mathcal{R})$. Checking $(d-1)(k+1) - (d-2)\cdot 1 = (d-1)k + 1$ and $(d-1)\cdot 2(k+1) - (d-2)\cdot 1 = 2(d-1)k + d$ shows that Theorem 2.5 takes one unified form across all $d \geq 2$:

$$\operatorname{srank}_{\mathbb{R}}(K) = C_d(\mathcal{R}) = (d-1)\dim_{\mathbb{R}} \mathcal{A}_\varphi - (d-2)\, m_{\text{irr}} \qquad (d \geq 2), \tag{9}$$

and the identical formula holds over $\mathbb{C}$ with $m_{\text{irr}} = |S|$ linear factors. At $d = 2$ (1-fold identity pairing), (9) reduces to $\dim_{\mathbb{K}} \mathcal{A}$; at $d = 3$ (2-fold bilinear multiplication), it coincides with the Winograd–Fiduccia–Zalcstein–Alder–Strassen complexity $2\dim_{\mathbb{K}} \mathcal{A} - m_{\text{irr}}$ [74, 23, 1, 15]; and for $d \geq 4$ ($(d-1)$-fold multiplication), it matches the multilinear Alder–Strassen lower bound of Bläser, Mayer, and Shringi [10] with equality. Notice the contrast with symmetric Waring rank: for a real binary form $F(u, v) = \operatorname{Re}((u + iv)^D)$ ($D \geq 2$), complex Waring rank is 2 whereas real symmetric Waring rank equals D [18, 61, 11]—a gap $D - 2$ that grows with degree. In (9),

complexifying $\mathbb{R} \hookrightarrow \mathbb{C}$ splits each irreducible real quadratic into two conjugate linear factors (m_{irr} increases by 1 per nonreal pair while $\dim \mathcal{A}$ is unchanged), producing a *degree-independent* real-over-complex penalty of $(d-2)(2-1) = d-2$ per conjugate pair because coordinatewise decompositions (1) break mode symmetry by placing $H_P(x_0^2; X)$ on mode 0 and $E_k(x_j^2; X)$ on modes $j \geq 1$.

3 Mathematical Proofs and Core Constructions

Radial $K(x) = f(\|x\|^2)$ on $\mathbb{R}^d$ $\xleftarrow[\text{surjection } s_j = x_j^2]{\text{Eq. (3)}}$ Hankel $(s_0, \dots, s_{d-1}) \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)$ on $\mathbb{R}^d_{\geq 0}$

Constructive upper bounds ($\leq R$)
- $\mathbb{C}$: $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ roots of unity (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{=}0$: Lagrange at $0, \dots, M^{\mathbb{C}}_{d,k}-1$ (Sec. 3.2)
- $\mathbb{R}$, $\mathrm{Im}\,\lambda{>}0$: $M^{\mathbb{R}}_{d,k} = 2(d-1)k+d$ roots (Prop. 2.9)

Necessity & multilinear lower bounds ($\geq R$)
- $d{=}2$ slice $\Rightarrow \dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty \Rightarrow T_t = e^{tA}$ (Sec. 3.1)
- Dual span of $v(\mathbb{R}_{\geq 0}) \Rightarrow$ linear $\ell_{n,j} \in V^*$ (Sec. 3.3)
- Alder–Strassen peeling + $+1$ shift (Sec. 3.3)

3.1 Necessity and the two-dimensional core

In two coordinates over a field $\mathbb{K}$, let $\mathrm{rowSpan}(K) \leq (X \to \mathbb{K})$ be the $\mathbb{K}$-linear span of the row sections $\{x \mapsto K(x,y) : y \in X\}$. Then $\mathrm{srank}_{\mathbb{K}}(K) = \dim_{\mathbb{K}} \mathrm{rowSpan}(K) \in \mathbb{N} \cup \{\infty\}$: if $K(x,y) = \sum_{n=0}^{R-1} f_n(x) g_n(y)$, every row $x \mapsto K(x,y)$ is a linear combination of $f_0, \dots, f_{R-1}$ so $\dim \mathrm{rowSpan}(K) \leq R$; conversely, if $b_0(x), \dots, b_{R-1}(x)$ is a basis of $\mathrm{rowSpan}(K)$, expanding each row $x \mapsto K(x,y) = \sum_{n=0}^{R-1} c_n(y) b_n(x)$ directly gives a rank-R sum of products. For the two-variable Hankel kernel $(s_0, s_1) \mapsto f(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is the *translate span*

$$\mathcal{W}(f) = \mathrm{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto f(u+t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ satisfies $\mathrm{srank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \mathrm{srank}_{\mathbb{R}}(K)$ in every dimension d. In $d = 2$, equality $\mathrm{srank}_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{srank}_{\mathbb{R}}(K)$ always holds: if real functions $b_0, \dots, b_{R-1} : X \to \mathbb{R}$ are $\mathbb{R}$-linearly independent, any complex relation $\sum_n (\alpha_n + i\beta_n) b_n(x) = 0$ forces $\sum_n \alpha_n b_n(x) = 0$ and $\sum_n \beta_n b_n(x) = 0$ separately, so $\dim_{\mathbb{C}} \mathrm{rowSpan}(K_{\mathbb{C}}) = \dim_{\mathbb{R}} \mathrm{rowSpan}(K)$.

For any $d \geq 2$, if $\mathrm{srank}_{\mathbb{R}}(x \in \mathbb{R}^d \mapsto \varphi(\|x\|^2)) < \infty$, the slice inequality (4) and 2-D complexification give $\dim_{\mathbb{C}} \mathcal{W}(\varphi_{\mathbb{C}}) < \infty$. Necessity and the 2-D annihilator theorem then follow from a single basis-free operator construction on $\mathcal{W}(\psi)$ for any continuous $\psi : \mathbb{R} \to \mathbb{C}$ with $\dim_{\mathbb{C}} \mathcal{W}(\psi) = n < \infty$:

1. Equip $\mathcal{W}(\psi)$ with a Banach norm transferred across a $\mathbb{C}$-linear isomorphism $\mathcal{W}(\psi) \simeq_{\mathbb{C}} \mathbb{C}^n$. Because all Hausdorff vector topologies on $\mathbb{C}^n$ coincide, convergence in this norm is equivalent to convergence at every point $u \in \mathbb{R}_{\geq 0}$ inside the fixed n-dimensional subspace $\mathcal{W}(\psi)$.

2. Non-negative shifts $(T_t g)(u) = g(u+t)$ $(t \geq 0)$ map $\mathcal{W}(\psi)$ into itself because shifting a translate $u \mapsto \psi(u+s)$ gives $u \mapsto \psi(u+(s+t)) \in \mathcal{W}(\psi)$. Continuity of ψ on $[0, \infty)$ implies that $t \mapsto (T_t g)(u)$ is continuous at each point $u \geq 0$, hence norm-continuous on $\mathcal{W}(\psi)$, and therefore $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ is continuous in operator norm on $[0, \infty)$.

3. By the Banach-algebra semigroup theorem (Section 4), there exists a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = (T_t \psi|_{\mathbb{R}_{\geq 0}})(0) = \mathrm{eval}_0(\exp(tA)\psi|_{\mathbb{R}_{\geq 0}})$.

4. Over the algebraically closed field $\mathbb{C}$, $\mathcal{W}(\psi) = \bigoplus_{\mu \in S} \ker((A - \mu I)^n)$ decomposes into generalized eigenspaces. On a summand where $(A - \mu I)^n v = 0$, writing $tA = \mu t I + t(A - \mu I)$ as a sum of commuting operators gives $\exp(tA)v = e^{\mu t} \sum_{q=0}^{n-1} \frac{t^q}{q!} (A - \mu I)^q v$, because all powers $q \geq n$ vanish. Evaluating at 0 turns each vector $(A - \mu I)^q v$ into a complex scalar, yielding $\psi(t) = \sum_{\mu \in S} P_\mu(t) e^{\mu t}$ on $[0, \infty)$. When $\psi = \varphi_{\mathbb{C}}$ for a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi_{\mathbb{C}}(t) + \overline{\varphi_{\mathbb{C}}(t)})$ and grouping conjugate rates $(\mu, \bar{\mu})$ produces a canonical real representation $\mathcal{R}$.

Finally, differentiating $t \mapsto \exp(tA)\psi|_{\mathbb{R}_{\geq 0}}$ q times on $[0,\infty)$ brings down A^q, so $q(D)\psi(t) = \mathrm{eval}_0\big(q(A)\exp(tA)\psi|_{\mathbb{R}_{\geq 0}}\big)$ for every polynomial $q \in \mathbb{C}[X]$. In $d = 2$, the Cayley–Hamilton theorem ($q = \chi_A \implies \chi_A(A) = 0$) immediately implies that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ satisfies $\chi_A(D)\psi = 0$ on $[0,\infty)$. Conversely, applying $(D-\mu)^{\deg P_\mu+1}$ kills $P_\mu(t)e^{\mu t}$ while shifting every other rate $\nu \neq \mu$ by an invertible factor $(\nu-\mu)^{\deg P_\mu+1}$ on leading terms, so $q(D)\big(\sum_{\mu\in S} P_\mu(t)e^{\mu t}\big) = 0$ on $[0,\infty)$ ($P_\mu \neq 0$) if and only if $p_{\min}(X) = \prod_{\mu\in S}(X-\mu)^{\deg P_\mu+1}$ divides $q(X)$. Hence $p_{\min} \mid \chi_A$, giving $\sum_{\mu\in S}(\deg P_\mu + 1) = \deg p_{\min} \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, matched by Taylor expansion of $P_\mu(s_0+s_1)e^{\mu(s_0+s_1)}$ to prove Theorem 2.6.

3.2 Constructive multi-way upper bounds via Hasse polynomials and roots of unity

Warm-up ($d = 3, k = 1$): extracting a degree-1 radial polynomial as an $[X^1]$ coefficient. Consider $P(s) = p_0 + p_1 s$ on $s = s_0 + s_1 + s_2$. Notice that if we put $U_0(X) = p_1 + (p_0 + p_1 s_0)X$, $U_1(X) = 1 + s_1 X$, and $U_2(X) = 1 + s_2 X$, then the coefficient of X^1 in $U_0(X)U_1(X)U_2(X)$ is

$$[X^1]\big((p_1+(p_0+p_1s_0)X)(1+s_1X)(1+s_2X)\big) = (p_0+p_1s_0)\cdot 1\cdot 1 + p_1s_1\cdot 1 + p_1\cdot 1\cdot s_2 = P(s_0+s_1+s_2),$$

a product of three univariate polynomials in X whose j-th factor depends *only on* s_j! Since $\deg(U_0U_1U_2) \leq 3$, over $\mathbb{C}$ averaging $\zeta^{-r}U_0(\zeta^r)U_1(\zeta^r)U_2(\zeta^r)$ over the $M^{\mathbb{C}}_{3,1} = 3$ cube roots of unity ($\zeta = e^{2\pi i/3}$) isolates the X^1 term (because X^0, X^2, X^3 shift to X^{-1}, X^1, X^2, none of which is a multiple of 3!), whereas over $\mathbb{R}$ for a nonreal rate $e^{\lambda(s_0+s_1+s_2)}$ ($\mathrm{Im}\,\lambda > 0$), Proposition 2.9 extracts $\mathrm{Re}([X^1]\prod_{j=0}^{2} e^{\lambda s_j}U_j(X))$ using $M^{\mathbb{R}}_{3,1} = 2(3-1)(1)+3 = 7$ real 3-way products.

General degree $k = \deg P$ ($d \geq 2$). Using the q-th Hasse derivative $D^{[q]}P(s_0) = \frac{1}{q!}P^{(q)}(s_0)$—simply the q-th Taylor coefficient of P around s_0, defined algebraically without division by $P(s_0+t) = \sum_{q=0}^{k}(D^{[q]}P)(s_0)\,t^q$—define the *reversed Hasse polynomial* $H_P(s_0;X)$ on mode 0 and the *truncated exponential polynomials* $E_k(s_j;X)$ on modes $j = 1,\dots,d-1$:

$$H_P(s_0;X) = \sum_{q=0}^{k} q!\,(D^{[q]}P)(s_0)X^{k-q},\quad E_k(s_j;X) = \sum_{q=0}^{k}\frac{s_j^q}{q!}X^q,\quad Q(X) = H_P(s_0;X)\prod_{j=1}^{d-1}E_k(s_j;X). \tag{10}$$

Binomial convolution gives $[X^q]\prod_{j=1}^{d-1}E_k(s_j;X) = (s_1+\cdots+s_{d-1})^q/q!$ for all $0 \leq q \leq k$, so $[X^k]Q(X) = P(s_0+\cdots+s_{d-1})$ with $\deg Q \leq dk$. Three coefficient-extraction schemes then yield all three upper bounds:

1. *Complex rates over $\mathbb{C}$:* Put $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ and $\zeta = e^{2\pi i/M^{\mathbb{C}}_{d,k}}$. Because $d \geq 2$, every monomial index $n \in \{0,\dots,dk\}$ satisfies $-M^{\mathbb{C}}_{d,k} < -k \leq n-k \leq (d-1)k < M^{\mathbb{C}}_{d,k}$, so $M^{\mathbb{C}}_{d,k} \mid (n-k) \iff n = k$. Discrete Fourier averaging $[X^k]Q(X) = \frac{1}{M^{\mathbb{C}}_{d,k}}\sum_{r=0}^{M^{\mathbb{C}}_{d,k}-1}\zeta^{-rk}Q(\zeta^r)$ multiplied by $\prod_{j=0}^{d-1}e^{\mu s_j}$ gives $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ complex products.
2. *Real rates over $\mathbb{R}$ ($\mathrm{Im}\,\lambda = 0$, $P \in \mathbb{R}[X]$):* On modes $1,\dots,d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1}E_k(s_j;X) \in \mathbb{R}[X]$ has $\deg A \leq (d-1)k < M^{\mathbb{C}}_{d,k}$. Real Lagrange interpolation of $A(X)$ at the $M^{\mathbb{C}}_{d,k}$ distinct real nodes $r = 0,1,\dots,M^{\mathbb{C}}_{d,k}-1 \in \mathbb{R}$ expresses each $[X^q]A(X)$ ($0 \leq q \leq k$), and hence $P(\sum_j s_j)e^{\lambda\sum_j s_j}$, as a sum of $M^{\mathbb{C}}_{d,k} = (d-1)k+1$ real d-way products.
3. *Nonreal rates over $\mathbb{R}$ ($\mathrm{Im}\,\lambda > 0$, Proof of Propositions 2.8 and 2.9):* Put $M = M^{\mathbb{R}}_{d,k} = 2(d-1)k + d = (2k+1)d - 2k$ (note $M = d$ when $k = 0$). Writing $\mathrm{Re}\,w = \frac{1}{2}(\bar{w} +$

w) in (8) and expanding over subsets $S \subseteq \{0,\dots,d-1\}$ of cardinality $m = |S|$ and degree tuples $q \in \{0,\dots,k\}^d$, the identity $(-1)^r = e^{i\pi r M/M}$ produces the geometric phase $\exp(2\pi i r K(S,q)/M)$ with integer frequency

$$K(S,q) = (2k+1)m + \sum_{j\notin S} q_j - \sum_{j\in S} q_j - k \in \mathbb{Z}.$$

Because $0 \le q_j \le k$ and $d \ge 2$: if $m = 0$ ($S = \emptyset$), $-M < -k \le K(\emptyset,q) = \sum_j q_j - k \le (d-1)k < M$, so $M \mid K \iff \sum_j q_j = k$; if $m = d$ ($S = \{0,\dots,d-1\}$), $K = M + (k - \sum_j q_j)$, so $M \mid K \iff \sum_j q_j = k$; and if $1 \le m \le d-1$,

$$1 \le (k+1)m - k \le K(S,q) \le (k+1)m + (d-1)k \le 2(d-1)k + d - 1 = M - 1,$$

so $M \nmid K(S,q)$ and $\sum_{r=0}^{M-1} e^{2\pi i r K(S,q)/M} = 0$. Applying (8) to $U_0(X) = e^{\lambda s_0} H_P(s_0; X)$ and $U_j(X) = e^{\lambda s_j} E_k(s_j; X)$ ($j \ge 1$) yields $M_{d,k}^{\mathbb{R}} = 2(d-1)k + d$ real d-way products.

3.3 Multilinear lower bounds via Alder–Strassen degree peeling

Step 1: From set-theoretic 1-D functions to multilinear dual functionals. Let V be a finite-dimensional $\mathbb{K}$-vector space ($\mathbb{K} \in \{\mathbb{R},\mathbb{C}\}$), let $v : \mathbb{R}_{\ge 0} \to V$ be an evaluation curve whose image spans V ($\mathrm{span}_{\mathbb{K}}(v(\mathbb{R}_{\ge 0})) = V$, verified by checking $\Lambda \circ v = 0 \implies \Lambda = 0$ for all $\Lambda \in V^*$), and let $\mathcal{M} : V^d \to \mathbb{K}$ be d-multilinear. If the Hankel kernel $(s_0,\dots,s_{d-1}) \mapsto \mathcal{M}(v(s_0),\dots,v(s_{d-1}))$ admits a rank-R decomposition (1) with arbitrary functions $f_{n,j} : \mathbb{R}_{\ge 0} \to \mathbb{K}$, choosing a basis $\{v(t_b)\}_{b=1}^{\dim V}$ of V and defining unique linear functionals $\ell_{n,j} \in V^*$ by $\ell_{n,j}(v(t_b)) = f_{n,j}(t_b)$ yields

$$\mathcal{M}(u^{(0)},\dots,u^{(d-1)}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u^{(j)}) \qquad \text{for all } u^{(0)},\dots,u^{(d-1)} \in V, \tag{11}$$

because both sides are d-multilinear maps agreeing on every d-tuple of basis vectors.

Step 2: Geometric intuition on $d = 3, k = 1$ ($\|x\|^2\cos(\|x\|^2)$ requires 7 real terms). Before stating the general induction, consider $K_3(x) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$, where $v(s) = (e^{is}, se^{is}) \in \mathbb{C}^2$ spans $V = \mathbb{C} \oplus \mathbb{C}\varepsilon \cong \mathbb{C}[T]/(T^2)$ ($\varepsilon^2 = 0$) over $\mathbb{R}$ and (11) represents the real 3-multilinear multiplication map $\mathcal{M}((u_0,u_1),(v_0,v_1),(w_0,w_1)) = \mathrm{Re}(u_1v_0w_0 + u_0v_1w_0 + u_0v_0w_1)$. Why must $R \ge 7$, even though difference quotients of two complex Gaussians have rank 6 for every $h \ne 0$?

- On the nilpotent subspace $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, $\mathcal{M}((0,z_1),(v_0,0),(w_0,0)) = \mathrm{Re}(z_1v_0w_0)$ vanishes only if $z_1 = 0$, so the restrictions $\{\ell_{n,0}|_E : 0 \le n < R\}$ must span E^* ($\dim_{\mathbb{R}} E^* = 2$). Picking two functionals (say $n = 0, 1$) forming a basis of E^*, there is a unique linear map $\gamma_0 : \mathbb{C} \to \mathbb{C}$ such that $\psi_0(u_0) := (u_0, \gamma_0(u_0)) \in \ker \ell_{0,0} \cap \ker \ell_{1,0}$ for all $u_0 \in \mathbb{C}$, *killing two terms* ($n = 0, 1$).
- Crucially, because $\varepsilon^2 = 0$ in $\mathbb{C}[T]/(T^2)$, substituting $\psi_0(u_0)$ on mode 0 and testing $(0,v_1) \in E$ on mode 1 gives $\mathcal{M}(\psi_0(u_0),(0,v_1),(w_0,0)) = \mathrm{Re}(u_0v_1w_0)$—*completely unaffected* by $\gamma_0(u_0)$! Thus the remaining functionals $\{\ell_{n,1}|_E : n \ge 2\}$ on mode 1 must still span E^*, letting a second map $\psi_1(v_0) = (v_0, \gamma_1(v_0))$ kill *two more terms* ($n = 2, 3$).
- Finally, evaluating $(u_0,v_0,w_1) \mapsto \mathcal{M}(\psi_0(u_0),\psi_1(v_0),(0,w_1)) = \mathrm{Re}(u_0v_0w_1)$ uses only the surviving $R - 4$ terms ($n \ge 4$). Since every $\mathbb{R}$-linear functional $\Lambda : \mathbb{C} \to \mathbb{R}$ has a nonzero kernel vector ($\dim_{\mathbb{R}} \mathbb{C} = 2 > 1$), induction on d shows $\mathrm{Re}(u_0v_0w_1)$ requires at least 3 real terms, forcing $R - 4 \ge 3$, i.e. $R \ge 7$.

Step 3: General multi-mode degree peeling over $\mathbb{C}$ and over $\mathbb{R}$. Over $\mathbb{C}$, let $V = \prod_{i\in\iota} \mathbb{C}^{\tau_i^0}$ with $\tau_i^0 = \deg P_i + 1$ and $v(s)_{i,q} = \frac{s^q}{q!} e^{\mu_i s}$. Exponential-polynomial uniqueness implies $\mathrm{span}_{\mathbb{C}}(v(\mathbb{R}_{\geq 0})) = V$, mapping the Hankel kernel to the truncated-convolution form with cutoffs $\tau_i \leq \tau_i^0$:

$$\mathcal{T}_\tau(u^{(0)}, \ldots, u^{(d-1)}) = \sum_{i\in\iota} \sum_{q_0+\cdots+q_{d-1}<\tau_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u_{i,q_j}^{(j)}, \qquad w_i(\tau_i - 1) \neq 0 \ (\tau_i > 0).$$

Strong induction on $\sum_i \tau_i$ peels one block i_0 at a time:

- When $\tau_{i_0} = 1$, testing $(e_{i_0,0}, \ldots, e_{i_0,0})$ finds $\ell_{n_0,0}(e_{i_0,0}) \neq 0$; projecting block i_0 to 0 on modes $j \geq 1$ and subtracting $\frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 kills 1 term and reduces τ_{i_0} to 0.
- When $\tau_{i_0} \geq 2$, since $2(\tau_{i_0} - 1) \geq \tau_{i_0}$, rank-1 perturbations along $e_{i_0,\tau_{i_0}-1}$ on modes $j = 0, \ldots, d-2$ do not cross-interact when another coordinate equals $e_{i_0,\tau_{i_0}-1}$, killing $d-1$ distinct terms; truncating modes $0, \ldots, d-2$ to $q < \tau_{i_0} - 1$ and shifting mode $d-1$ up by $+1$ $(e_{i_0,q} \mapsto e_{i_0,q+1})$ reduces $\tau_{i_0} \mapsto \tau_{i_0} - 1$ with shifted top weight $w'_{i_0}(\tau_{i_0} - 2) = w_{i_0}(\tau_{i_0} - 1) \neq 0$.

Over $\mathbb{R}$, the same induction on $G = (\prod_{i\in\iota_{\mathbb{R}}} \mathbb{R}^{\tau^0_{\mathbb{R},i}}) \times (\prod_{u\in\iota_{\mathbb{C}}} \mathbb{C}^{\tau^0_{\mathbb{C},u}})$ peels real blocks identically (1 term at $\tau_{\mathbb{R},i_0} = 1$, $d-1$ terms at $\tau_{\mathbb{R},i_0} \geq 2$), peels nonreal $\tau_{\mathbb{C},u_0} = 1$ blocks by d terms (using $\dim_{\mathbb{R}} \ker(\Lambda : \mathbb{C} \to \mathbb{R}) \geq 1$ to kill 2 terms on mode 0 and 1 term on each mode $1, \ldots, d-2$), and peels nonreal $\tau_{\mathbb{C},u_0} \geq 2$ blocks by $2(d-1)$ terms (2 independent real functionals on $0 \oplus \mathbb{C}e_{u_0,\tau_{\mathbb{C},u_0}-1} \cong \mathbb{R}^2$ per mode $j = 0, \ldots, d-2$, followed by $+1$ index shifting on mode $d-1$), completing Theorem 2.5.

3.4 Proof of the general multi-Gaussian border-rank bound (Theorem 2.7)

Identity (6) is immediate from (5): for $d \geq 2$, $(k+1) + (d-2)k = (d-1)k + 1$ and $d(k+1) + (d-2)k = 2(d-1)k + d$. For a single well-formed component (λ, P) of degree $k = \deg P$ and any $h \in \mathbb{R} \setminus \{0\}$, replace $s = \sum_{j=0}^{d-1} x_j^2$ inside $P(s)$ by the exponential difference quotient $u_h(s) = \frac{e^{hs}-1}{h}$ (setting $u_0(s) := 0$ via $0^{-1} = 0$ in Lean):

$$F_h^{(\lambda,P)}(x) := \mathrm{Re}\Big(P\big(u_h(\|x\|^2)\big)\, e^{\lambda\|x\|^2}\Big).$$

Because $u_h(s) = L_h(e^{hs})$ is a degree-1 polynomial $L_h(Y) = -h^{-1} + h^{-1}Y$ in $Y = e^{hs}$, the composed polynomial $Q_h(Y) = (P \circ L_h)(Y)$ still has degree at most k:

$$P(u_h(s))\, e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h)\, (e^{hs})^m e^{\lambda s} = \sum_{m=0}^{k} ([Y^m]Q_h) \prod_{j=0}^{d-1} e^{(\lambda+mh)x_j^2}.$$

When $\mathrm{Im}\,\lambda = 0$ (so $P, Q_h \in \mathbb{R}[X]$), this is a sum of $k+1 = b_d(\lambda, P)$ real Gaussian products; when $\mathrm{Im}\,\lambda > 0$, it is a sum of $k+1$ complex Gaussian products, whose real part has real d-way rank at most $d(k+1) = b_d(\lambda, P)$ by Proposition 2.8. (At $k = 1$ and $\lambda = 0$, $u_h(s) = h^{-1}(\prod_{j=0}^{d-1} e^{hx_j^2} - 1)$ realizes the classical W-tensor degeneration $[X^1]\prod_{j=0}^{d-1}(1 + s_j X)$ of exact rank d and border rank 2 [8, 14, 78] directly by smooth radial Gaussians.) Summing over $(\lambda_u, P_u) \in \mathcal{R}$ gives $\mathrm{srank}_{\mathbb{R}}(F_h) \leq B_d(\mathcal{R})$ for all $h \in \mathbb{R}$. Finally, for any compact $C \subset \mathbb{R}^d$, let $B = \sup_{x\in C} \|x\|^2 < \infty$. Twice applying the mean-value inequality to $y \mapsto e^y - 1 - y$ gives $|e^y - 1 - y| \leq y^2 e^{|y|}$, so $|u_h(s) - s| \leq |h|(B+1)^2 e^{B+1} \to 0$ uniformly for $s \in [0, B]$ as $h \to 0$ $(h \neq 0)$. Uniform continuity of $(u, s) \mapsto \mathrm{Re}(P(u)e^{\lambda s})$ on $[-1, B+1] \times [0, B]$ then implies $F_h \to \mathrm{profile}(\mathcal{R}, \|\cdot\|^2)$ uniformly on C.

4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem [52, 22]: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s+t) = M(s)M(t)$ for all $s, t \geq 0$, and M is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \geq 0$.

4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

Integral smoothing (`Semigroup.lean`). Following Nagumo [52] (see Engel and Nagel [22], Thm. I.3.7), since $M(0) = 1$ and M is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\,du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series. For $t \geq 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\,du$ shows by the fundamental theorem of calculus that M is right-differentiable on $[0, \infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`). Without integration, if $\|M(u) - 1\| \leq \frac{1}{2}$ on $[0, \delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_n) - 1)/h_n \to A$; along h_n, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

Nonstandard analysis with homogeneous common scale. Euler's heuristic sets $A = \mathrm{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$ (compare nonstandard treatments of Lie's theorems and Hilbert's fifth problem [33, 28]). Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{12}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0, 1]$ and $\left\|\frac{x}{\eta}\right\| \leq 1$, both rescaled quantities are bounded by 1 *without any analytical estimate*. In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \mathrm{st}(x/\eta)$ with $\|B\| \leq 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0, 1]$. Comparing $N_{\mathrm{step}} = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N_{\mathrm{step}}\varepsilon \approx sa$) against N_{step} steps of $\exp(\eta B)$ (arriving at time $N_{\mathrm{step}}\eta \approx s$) for an arbitrary standard $s \geq 0$,

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \leq \left\|\frac{x}{\eta} - B\right\| + \left\|B - \frac{\exp(\eta B) - 1}{\eta}\right\| \approx 0,$$

so summing N_{step} single-step errors of size $\eta \cdot o(1)$ with $N_{\mathrm{step}}\eta \leq s$ yields $M(sa) = \exp(sB)$ for all standard $s \geq 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \geq 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$. Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

4.2 Three nonstandard formalizations and quantitative comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

Table 3: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$. Every project helper lemma in each proof's module dependencies is counted in full (excluding standard Mathlib and Isabelle/HOL library imports).

Proof method	Prover / files	Phys.	Nonbl.	Scope
Integral smoothing (main)	Lean 4 (`Semigroup.lean`)	96	87	Any Banach algebra
Finite differences	Lean 4 (`GeneratorBound`, `FD`)	256	233	Finite-dim.
NSA (ultrafilter seq.)	Lean 4 (`NonstandardSemigroup`)	206	190	Finite-dim.
NSA (ultrapower API)	Lean 4 (`Ultrapower*` + shared)	399	344	Finite-dim.
NSA (star + `transfer`)	Isabelle (`Semigroup_NSA.thy`)	275	253	Finite-dim. ($\|1\| = 1$)

1. **Isabelle/HOL (`Isabelle/Semigroup_NSA.thy`):** Built on Fleuriot's nonstandard analysis development [24], Hölzl, Immler, and Huffman's typeclass hierarchy [35] (see also Unruh and Rodríguez Caballero [70] for recent operator theory in Isabelle/HOL), and Huffman's nonstandard `transfer` tactic (`transfer_principle.ML` in `HOL-Nonstandard_Analysis`). Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \le e^h$ with $h \ge 0$, then $\|u^n - v^n\| \le ne^{nh}\|u - v\|$. Since $\|M(\varepsilon)\| \le 1 + \|x\| \le 1 + \eta \le e^\eta$ and $\|\exp(\eta B)\| \le e^{\eta\|B\|} \le e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps via compact-interval bounds instead.)
2. **Lean 4 with ultrafilter sequences (`Experiments/NonstandardSemigroup.lean`):** Expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining B in one line from compactness of the closed unit ball along ultrafilters.
3. **Lean 4 in nonstandard vocabulary (`Experiments/Ultrapower*.lean`):** Builds a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\mathrm{h}}$, one-variable transfer lemmas) and writes Euler's proof in classical nonstandard notation.

Table 3 summarizes the exact line counts. Two caveats are essential when interpreting these numbers. First, line counts measure *library amortization* at a specific snapshot (`Mathlib v4.36.0-rc2` vs. `Isabelle2026`), not intrinsic mathematical complexity: `Semigroup.lean` (96 lines) is shortest and holds in every Banach algebra because Mathlib [49, 21] already packages Bochner integration, Neumann series, and the fundamental theorem of calculus into one-line lemmas, whereas compactness-based proofs are inherently finite-dimensional. Second, single-author comparisons carry *author-fluency bias*: standard Lean/Mathlib analysis is far better represented in AI pre-training corpora than Isabelle's `HOL-Nonstandard_Analysis`, so a specialist in Isabelle's NSA tactics might shorten `Semigroup_NSA.thy` further. Subject to those caveats, within Lean itself the 2× gap between integral smoothing (96 lines) and ultrafilter sequences (206 lines) persisted even after eliminating the 124-line difference-quotient bound via homogeneous coordinates (12), and discounting the 195-line interface `Ultrapower.lean` leaves $169 + 35 = 204$ lines (confirming that without automated Łoś transfer, nonstandard vocabulary in Lean merely wraps filter limits). In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, while 59 lines of local helpers (vector standard parts and half-line continuity via $M(\max(t, 0))$) and dual standard/nonstandard type statements bring the total to 275 lines.

Table 4: Module architecture and verified source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, exact & border costs	253	215
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; half-line shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; canonical conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient differential annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_\mu((d-1)\deg P_\mu + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D/2-D theorems & general border-rank bound	538	516
Total (`KernelRankLean`)		**5,750**	**5,432**

5 Formalization Architecture and Verification Audit

Table 4 breaks down the `KernelRankLean` library in Lean 4 [51, 49] (5,750 physical lines, 5,432 nonblank lines; earlier commits comprised 5,578 lines at `1094bf1`, 5,324 lines at `9722ff5`, 3,822 lines at `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at `240dea0` for 2-D classification alone). Whereas prior formalized tensor-rank developments establish hierarchical expressiveness in Isabelle/HOL [4] or finite-dimensional octonion and $\mathbb{F}_2$ matrix-multiplication rank bounds in Lean 4 [37, 75], our library couples continuous-domain Banach-algebra analysis with exact arbitrary-dimension real and complex multilinear rank formulas. Three engineering choices proved decisive:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a dependent family $\prod_{i\in\iota} X_i$ unifies the 2-D and d-D theorems, while specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (via a pattern-matched helper `fin2Vec`) avoids higher-order unification timeouts on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.
2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting finite-dimensional linear maps inherit automatic continuity.
3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 2.6 directly.

Constructive witnesses, `noncomputable` definitions, and axioms. In Lean 4, `separableRank` is marked `noncomputable` because taking the infimum in Definition 2.1 over all univariate factor families $X_i \to \mathbb{K}$ has no general decision procedure, and Mathlib's real and complex numbers are classical quotient types. Nevertheless, all upper-bound and border-rank proofs in `MultiComplexRank.lean`, `MultiRealRank.lean`, and `Main.lean` construct *explicit closed-form* univariate factor formulas (Propositions 2.8–2.9 and Section 3.4), evaluated directly in `Applications/acoustics_3d.py`. Across all 22 audited top-level theorems, `#print axioms` reports only Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), entering through Mathlib's Cauchy/quotient reals, coordinate surjection right inverses (2), finite-dimensional basis extraction in $\mathcal{W}(\psi)$ and V, and ultrafilter existence in `Experiments/`.

Automated verification audit. A single script (`scripts/check`, wired as the repository's pre-commit hook) builds every module of `KernelRankLean` and `Experiments` via `lake build`, verifies zero `sorry`s in compiler output, runs `#print axioms` on all 22 top-level Lean 4 theorems, builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

6 Analytical and GPU Applications

6.1 Experimental platform and Fourier rank invariance

All numerical optimizations, finite-precision simulations, and GPU benchmarks in Sections 6.3–6.4 are implemented in reproducible scripts under `Applications/` (with explicit coefficients in Appendix A) and executed on a workstation running Linux `7.0.0-43-generic` (`x86_64`) with an Intel Core Ultra 5 125H CPU (14 cores / 18 threads), 61 GiB system RAM, and an NVIDIA GeForce RTX 4070 Ti Super GPU (16,376 MiB GDDR6X, 672 GB/s peak bandwidth, driver `615.71.09`), using Python `3.14.4`, NumPy `2.5.4`, SciPy `1.18.1` (HiGHS dual-simplex LP solver), PyTorch `2.15.0` (CUDA `13.2`), and Triton `3.9.0`.

Before turning to discrete designs, note a continuous-domain Fourier duality: for $\operatorname{Re}\lambda < 0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi) = (\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ $(\deg P = k)$ to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P} = k$ and $\operatorname{Re}(1/(4\lambda)) < 0$. By Theorems 2.4 and 2.5, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

6.2 Harmonic "beach-ball" differential operators and steerable kernels

Complex partial-derivative operators $(\partial_x + i\partial_y)^m$ provide a canonical rotation-equivariant basis for 2-D symmetry and steerable filters [26, 40, 7, 60]. Specializing Proposition 2.8 to equal symbolic factors $z_0 = \cdots = z_{m-1} = \partial_x + i\partial_y$ (writing $m \geq 1$ for the harmonic order and $D_\theta = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ for the directional derivative along angle θ) gives, for *every* integer $m \geq 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m}\sum_{r=0}^{m-1}(-1)^r D^m_{\pi r/m}. \tag{13}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ $(n = 0,\dots,2m-1)$ of radius $h \to 0$, antipodal points satisfy $D_{\theta_{r+m}} = -D_{\theta_r}$ while Fourier orthogonality of $(-1)^r = e^{i\pi r}$ annihilates all Taylor orders below m. For $f \in C^m$ near (x,y), (13) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order $(m = 2N+1)$ hexagonal/decagonal "beach-ball" operators of Niemitalo [56] and the even-order operators of Perona [59] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x + h\cos\frac{\pi n}{m},\, y + h\sin\frac{\pi n}{m}\Big) = \frac{1}{(m-1)!\,2^{m-2}}\operatorname{Re}\big((\partial_x + i\partial_y)^m\big)f(x,y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x + i\partial_y)^m)$. Recognizing $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial^m_{\bar z}$ to *any* smooth radial kernel $\varphi(x^2+y^2) = \varphi(z\bar z)$ produces no lower-order radial derivatives [60]: since $\partial_{\bar z}(z) = 0$, induction gives

$$\left(\frac{\partial_x + i\partial_y}{2}\right)^m \varphi(x^2+y^2) = (x+iy)^m\,\varphi^{(m)}(x^2+y^2) = r^m e^{im\theta}\,\varphi^{(m)}(r^2). \tag{14}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with n_{re} real and n_{cx} nonreal rates (all $\lambda_u \neq 0$, approximating, e.g., an ideal circular low-pass filter [57]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_u^m) of continuous 2-way rank $R = n_{\mathrm{re}} + 2n_{\mathrm{cx}}$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [55]—absorbs the monomials of $\mathrm{Re}((x+iy)^m)$ and $\mathrm{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2 \rfloor + 1\big)(n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \cos(m\theta), \qquad \lceil m/2 \rceil (n_{\mathrm{re}} + 2n_{\mathrm{cx}}) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms. On the 161×161 ($\omega_c = \pi/64$) orientation-estimation filter of [55], replacing direct $2 \times 161^2 = 51{,}842$ taps/pixel by a two-complex-Gaussian ($R = 4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce simultaneous evaluation of $(C_m * I, S_m * I)$ below separate evaluation $2(\lfloor m/2 \rfloor + 1)R + 2\lceil m/2 \rceil R = 2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into P_y shared vertical passes followed by $\mathrm{rank}(A_c) + \mathrm{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\mathrm{rank}(B_c) + \mathrm{rank}(B_s)$ vertical passes, then

$$\begin{aligned} \dim \mathrm{rowSpan}_x(C_m, S_m) &\leq \mathrm{rank}(A_c) + \mathrm{rank}(A_s) + P_x, \\ \dim \mathrm{colSpan}_y(C_m, S_m) &\leq P_y + \mathrm{rank}(B_c) + \mathrm{rank}(B_s). \end{aligned}$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0, \ldots, x^m\} g_n(x)$ in x and $\{y^0, \ldots, y^m\} g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m+1)R$ passes ($4R$ passes for (h_x, h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^q g_n(u)$, halving multiplications per sampled pass via $I[y+\ell] \pm I[y-\ell]$ folding; (ii) sharing $\{I *_y (y^q g_n)\}_{q=0}^{M}$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m = 0, 1$; $10R$ instead of $12R$ for $m = 1, 3$); and (iii) for the d-D gradient $(h_{x_0}, \ldots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d \geq 3$, base rank $R_d = n_{\mathrm{re}} + d\, n_{\mathrm{cx}}$), recursively splitting $\{0, \ldots, d-1\}$ into halves to reduce $d^2 R_d$ passes to $T(d)R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2 \rfloor) + T(\lceil d/2 \rceil) + d \leq d(\lceil \log_2 d \rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

6.3 Worked application I: 2-D circular lens blur and $\mathrm{GL}_R(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [54, 27, 58], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r \leq 1}$ ($r = \sqrt{x^2 + y^2}$), which has infinite separable rank: for any $0 < r_1 < \cdots < r_M < 1$, evaluating $\mathbf{1}_{x_i^2 + y_j^2 \leq 1}$ at $x_i = r_i$, $y_j = \sqrt{1 - r_j^2}$ produces an invertible $M \times M$ upper-triangular binary matrix. Introducing a transition band $[1, S]$ with $S = 1 + \Delta r$ ($\Delta r = 0.2$) and truncating each 1-D pass to $|x|, |y| \leq S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \leq \sqrt{2}\, S \approx 1.697$. For any fixed rates (α_u, ω_u), minimizing the sampled Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \leq 1}|$ (2,000 radial nodes) over all polynomial weights is a linear program, solved with HiGHS inside differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`.

Figure 2 and Table 5 highlight four structural findings:

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches $C_2(\mathcal{R})$ identically. Distinct complex Gaussians achieve geometric ripple decay with $R = 2n_{\mathrm{cx}}$ ($0.2324 \to 0.0019$ from $1c$ to $6c$, or 0.0014 under global Remez search [58]), while adding a single real Gaussian ($1c+1r$, $2c+1r$) interpolates cleanly

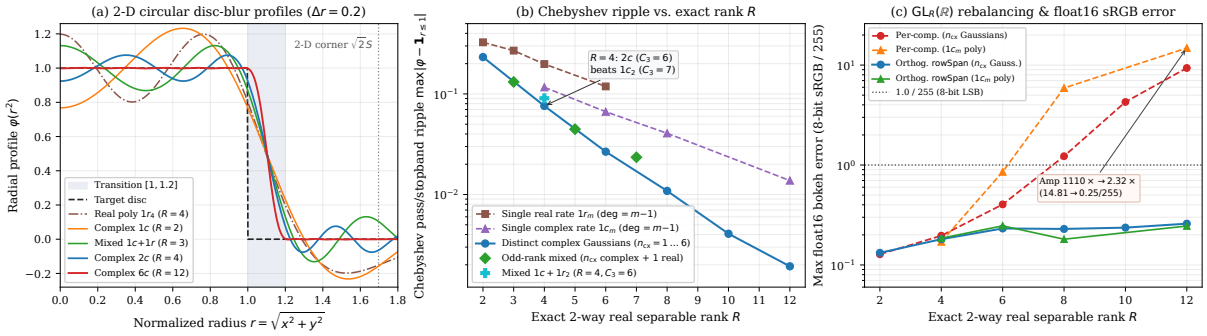


Figure 2: Two-dimensional circular disc blur ($\Delta r = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $R \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh error in 8-bit sRGB units (/255, via IEC 61966-2-1 sRGB encoding on a 512×512 linear-light HDR scene with night shadows 0.001 and 3×3 glints up to 48.0, $R_{\text{px}} = 25$, $W = 61$) before and after nuclear-norm-optimal GL$_R(\mathbb{R})$ rebalancing on rowSpan(K).

Table 5: Representative 2-D circular disc-blur designs ($\Delta r = 0.2$, $R_{\text{px}} = 25$, $W = 61$; explicit parameters in Table 9). Exact 2-D rank $R = C_2(\mathcal{R})$ and 3-D ball rank $C_3(\mathcal{R})$ follow Theorem 2.5; SVD is the empirical singular-value count ($\sigma_j > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error in 8-bit sRGB units (/255, IEC 61966-2-1).

Structure	Degrees	R (d=2)	SVD	C_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	sRGB raw	sRGB orth
$1c$ (Gaussian)	0	2	2	3	0.2324	1.65	1.30	0.13	0.13
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.16	0.20
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.20	0.18
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.12	0.23
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.17	0.19
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.21	0.23
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.26	0.20
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.22	0.23
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	9.35	0.26
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	14.81	0.25

at odd ranks $R = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $R = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($C_3 = 6$ vs. $C_3 = 7$ for $1c_2$ and $1r_4$).

2. **Transitive GL$_R(\mathbb{R})$ freedom in $d = 2$ vs. restricted CP orbits in $d \geq 3$:** While $R = \dim \text{rowSpan}(K)$ is fixed, in $d = 2$ writing the $W \times W$ symmetric kernel matrix as $K = HV^\top$ ($H, V \in \mathbb{R}^{W \times R}$) allows replacing (H, V) by $(HT, VT^{-\top})$ for *any* invertible matrix $T \in \text{GL}_R(\mathbb{R})$ [68], mixing basis vectors freely across distinct components. Splitting high-order fits component-by-component causes severe intermediate cancellation ($\kappa_\infty^{\text{raw}} = 595$ on $6c$ and 1,110 on $1c_6$), producing visible 9.35/255 and 14.81/255 8-bit sRGB errors (amplified in dark shadows around 48× HDR bokeh glints by the 12.92× linear toe of IEC 61966-2-1 sRGB) when intermediate passes are stored in IEEE `float16` (as in WebGPU `pack2x16float` [58]). To remove this cancellation without a full $W \times W$ SVD, compute a thin QR factorization $H = QR_H$ ($Q^\top Q = I_R$), project $K = QMQ^\top$ onto the tiny $R \times R$ symmetric matrix $M = R_H V^\top Q$, and diagonalize $M = U \,\text{diag}(e_1, \ldots, e_R) U^\top$

with orthogonal U. Setting $G = QU = [g_1, \ldots, g_R]$ ($g_m^\top g_n = \delta_{mn}$) yields the exact eigendecomposition $K(x,y) = \sum_{n=1}^{R} e_n g_n(x) g_n(y)$, where $|e_n|$ are the nonzero singular values of K; setting $\tilde{h}_n = \sqrt{|e_n|}\, g_n$ and $\tilde{v}_n = \operatorname{sign}(e_n)\tilde{h}_n$ (accumulated in ascending order of $|e_n|$) achieves the minimum possible ℓ_2 energy sum $\sum_{n=1}^{R} \|\tilde{h}_n\|_2 \|\tilde{v}_n\|_2 = \sum_{n=1}^{R} |e_n|$ (the *nuclear* or *trace norm* $\|K\|_{S_1}$), collapsing κ_∞ to ≤ 2.50 and keeping `float16` 8-bit sRGB max error at 0.09–0.33/255 (0.13–0.26/255 in Table 5; sRGB RMSE $\leq 0.021/255$, linear max error $\leq 0.20/255$) across all 18 tested families. By contrast, in $d \geq 3$, minimal d-way CP decompositions do not admit transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations [43], making well-separated rates and amplitude-balanced root-of-unity factors essential.

3. **GPU throughput vs. both direct spatial `conv2d` and global `cuFFT` (Table 7):** On 1920×1080 RGBA images, Table 7 compares two-pass symmetric Triton FIR kernels (clamped borders, constant two-buffer memory) against direct $W \times W$ PyTorch depthwise `conv2d` and global `torch.fft.rfft2/irfft2` (both zero-padded linear convolution and explicit L-pixel replicate-clamped padding matching the spatial border rule, with precomputed $\widehat{K}$ in `float32`). Separable Triton beats direct $W \times W$ `conv2d` by up to 35.6× at $R_{\mathrm{px}} = 64$, and in `float16` beats clamp-padded `cuFFT` at small-to-moderate radii and modest ranks ($R_{\mathrm{px}} = 8$ for $R \in \{4, 8\}$; $R_{\mathrm{px}} = 16$ for $R = 4$). For purely *shift-invariant* blurs at $R_{\mathrm{px}} \geq 32$ (or $R = 12$), global `cuFFT` (1.10–1.87 ms) is faster; separable decompositions remain advantageous at large radii primarily when (i) each 1-D complex Gaussian uses an $O(1)$-per-pixel recursive IIR filter [19, 76], (ii) the blur radius $R_{\mathrm{px}}(x, y)$ varies per pixel in depth-of-field shaders [27] where global FFTs do not apply, or (iii) graphics pipelines lack padded complex spectral buffers.

4. **Direct discrete rank-R optimization vs. sampled continuous profiles (Figure 3, Table 6):** Because any finite-tap separable cascade on $\{-L, \ldots, L\}^d$ is rectangular-truncated and even in every coordinate ($h_n[-m] = h_n[m]$), its 1-D factors lie in an $(L+1)$-dimensional even subspace, capping 2-D discrete rank at $\min(R, L+1)$ (for instance, sampling $6c$ of continuous rank 12 gives exact discrete SVD rank $4, 6, 11, 12, 12, 12$ at $R_{\mathrm{px}} = 2, 4, 8, 16, 25, 64$, with σ_{12}/σ_1 locking onto 4.37–4.34×10^{-4} for $R_{\mathrm{px}} \geq 16$). Does optimizing discrete rank-R matrices directly on $\{-L, \ldots, L\}^2$ improve on sampling continuous complex Gaussians? Table 6 and Figure 3 compare sampled continuous Gaussians against (i) direct Eckart–Young ℓ_2 SVD truncation of a raised-cosine smooth disc on $[1, 1.2]$, (ii) unconstrained D_4-symmetric discrete rank-R minimax optimization (alternating pass/stopband clamping with SVD rank-R projection), and (iii) shell-isotropic discrete rank-R minimax (additionally equalizing lattice points of equal Euclidean radius $m_x^2 + m_y^2 = N$, measuring shell anisotropy $\Delta_{\mathrm{sh}} = \max_N(\max_{\|\mathbf{m}\|^2=N} K - \min_{\|\mathbf{m}\|^2=N} K)$). Three regimes emerge clearly:

   - *Small stencils ($R_{\mathrm{px}} \leq 6$, $L+1 \leq 9$):* When R is comparable to $L+1$, direct discrete optimization exploits the tiny $(L+1)$-dimensional space (0.0000 error once $R \geq L+1$, and ±0.0048 vs. continuous ±0.0759 at $R_{\mathrm{px}} = 6, R = 4$; notice in $\{0, \ldots, 7\}^2$ only $1^2 + 7^2 = 5^2 + 5^2 = 50$ shares a squared radius, so even shell-isotropic discrete optimization retains 0°-vs.-45° angular freedom in Figure 3).
   - *Direct ℓ_2 SVD vs. Chebyshev minimax:* At $R_{\mathrm{px}} \geq 12$, ℓ_2 SVD truncation degrades in both pass/stopband ripple (0.0956–0.1030 vs. 0.0759 at $R = 4$) and severe 17.7%–19.2% equal-radius shell anisotropy, producing visible cross-streaks around the 12× HDR star in Figure 3.
   - *Anisotropy trade-off and large-stencil convergence ($R_{\mathrm{px}} \geq 25$):* Unconstrained discrete minimax lowers pass/stopband ripple (0.0523 at $R_{\mathrm{px}} = 25, R = 4$) *only by warping the unconstrained transition rim* $1 < r < 1.2$ *anisotropically* ($\Delta_{\mathrm{sh}} = 0.091$ at $R_{\mathrm{px}} = 25$, 0.158 at $R_{\mathrm{px}} = 12$; see the 0° blue vs. 45° red split and squarish $K = 0.5$ rim contour in

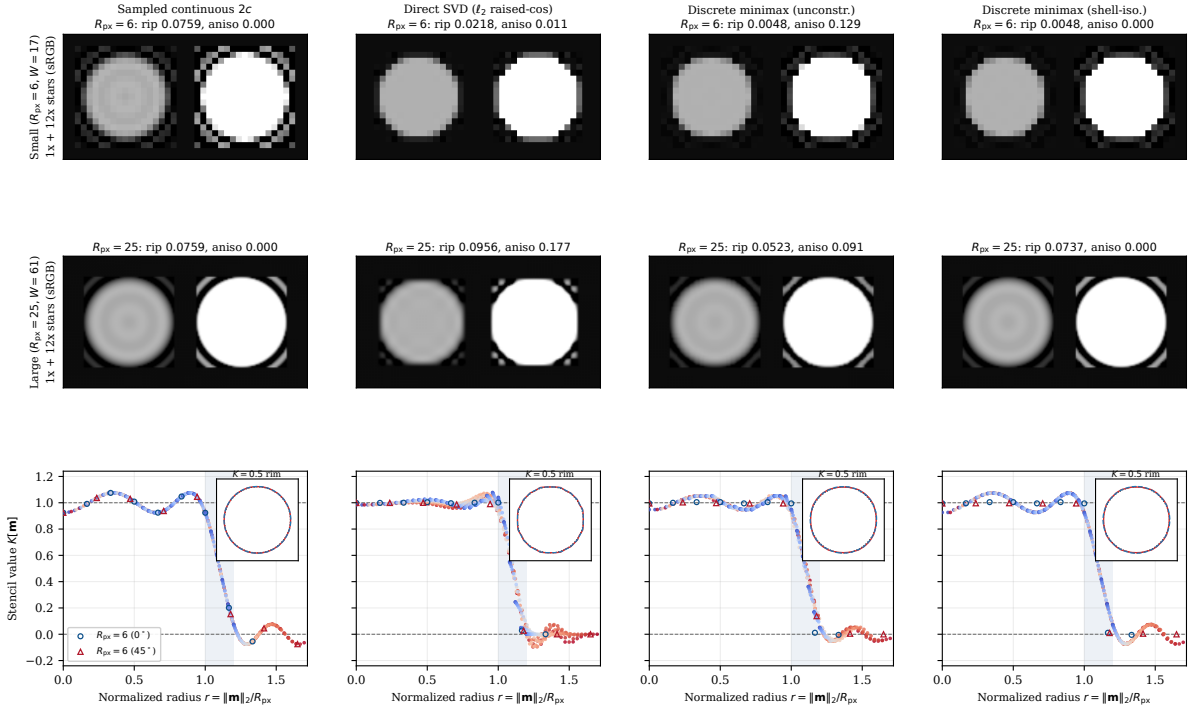


Figure 3: Visual and quantitative comparison at exact rank $R = 4$ between sampled continuous $2c$ Gaussians, direct ℓ_2 SVD of a raised-cosine disc, unconstrained discrete rank-4 minimax, and shell-isotropic discrete rank-4 minimax (`Applications/blur_2d.py`): top row, small stencil ($R_{\text{px}} = 6, W = 17$) sRGB bokeh on 1× (left) and 12× HDR (right) point stars; middle row, large stencil ($R_{\text{px}} = 25, W = 61$) sRGB bokeh; bottom row, per-pixel radial scatter ($R_{\text{px}} = 25$ dots colored from 0° blue to 45° red; $R_{\text{px}} = 6$ open 0°/45° markers) and $K = 0.5$ rim contours ($R_{\text{px}} = 6$ solid red, $R_{\text{px}} = 25$ dashed blue, reference circle dotted black).

Figure 3). Once equal-radius shell isotropy is enforced, discrete optimization converges right onto the continuous complex-Gaussian curve (0.2324 vs. 0.2324 at $R = 2$, 0.0758 vs. 0.0759 at $R = 4$, 0.0107 vs. 0.0109 at $R = 8$, and 0.0017 vs. 0.0019 at $R = 12$ at $R_{\text{px}} = 64$), without sacrificing continuous radius rescalability.

6.4 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{15}$$

Substituting a discrete plane wave $p^n[\mathbf{m}] = e^{i(\mathbf{k}\cdot\mathbf{m} - \omega_{\text{num}} n \Delta t)}$ with normalized wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$ turns the left-hand side of (15) into $2\cos(\omega_{\text{num}} \Delta t)\, p^n[\mathbf{m}]$ and the right-hand side into $\widehat{M}(\mathbf{k})\, p^n[\mathbf{m}]$, where $\widehat{M}(\mathbf{k}) = \sum_{\mathbf{m}} M[\mathbf{m}] e^{-i\mathbf{k}\cdot\mathbf{m}}$ is the discrete Fourier symbol of M. Since an exact continuous plane wave travels at speed c_0 ($\omega_{\text{exact}} \Delta t = c_0(\|\mathbf{k}\|_2/\Delta x)\Delta t = \nu\|\mathbf{k}\|_2$), matching the exact wave equation requires $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu\|\mathbf{k}\|_2)$ [65], and the numerical-to-exact phase-velocity ratio is $c_{\text{num}}(\mathbf{k})/c_0 = \arccos(\widehat{M}(\mathbf{k})/2)/(\nu\|\mathbf{k}\|_2)$. Whenever $|\widehat{M}(\mathbf{k})| \le 2$, both roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ have $|\zeta| = 1$ (spectral von Neumann stability); strict interior containment $\widehat{M}(\mathbf{k}) \in (-2, 2)$ for $\mathbf{k} \neq \mathbf{0}$ avoids repeated roots $\zeta = \pm 1$ (linear-in-n secular

Table 6: Sampled continuous complex Gaussians vs. direct discrete rank-R optimization across stencil radii $R_{\mathrm{px}} \in \{3, 6, 12, 25, 64\}$ ($L+1 = \lceil 1.2\, R_{\mathrm{px}} \rceil + 1$ even 1-D degrees of freedom). Each entry lists pass/stopband Chebyshev ripple $\pm\delta$, with equal-radius shell anisotropy Δ_{sh} in parentheses where nonzero.

R_{px}	W	$L{+}1$	R	Cont. Gaussian	Direct ℓ_2 SVD (Δ_{sh})	Disc. unconstr. (Δ_{sh})	Disc. shell-iso.
3	9	5	4	0.0759 (0)	0.0000 (0)	0.0000 (0)	0.0000
6	17	9	4	0.0759 (0)	0.0218 (0.011)	0.0048 (0.129)	0.0048
12	31	16	4	0.0759 (0)	0.0950 (0.107)	0.0190 (0.158)	0.0427
25	61	31	4	0.0759 (0)	0.0956 (0.177)	0.0523 (0.091)	0.0737
64	155	78	4	0.0759 (0)	0.1030 (0.192)	0.0665 (0.032)	0.0758
12	31	16	8	0.0109 (0)	0.0098 (0.014)	0.0014 (0.024)	0.0008
25	61	31	8	0.0109 (0)	0.0129 (0.015)	0.0049 (0.023)	0.0055
64	155	78	8	0.0109 (0)	0.0163 (0.019)	0.0076 (0.011)	0.0107
64	155	78	12	0.0019 (0)	0.0048 (0.006)	0.0010 (0.007)	0.0017

Table 7: GPU execution times (milliseconds) for 1920×1080 RGBA 2-D blur on an NVIDIA RTX 4070 Ti Super across disc radii $R_{\mathrm{px}} \in \{8, 16, 32, 64\}$ ($W = 2L{+}1$, $L = \lceil 1.2\, R_{\mathrm{px}} \rceil$). Columns compare replicate-clamped direct $W \times W$ depthwise `conv2d` (`fp32`), global `cuFFT` `rfft2`/`irfft2` with precomputed $\widehat{K}$ (`fp32`, both zero-padded linear convolution and L-pixel replicate-clamped borders), and two-pass symmetric Triton FIR cascades at ranks $R \in \{4, 8, 12\}$ (`fp32` / `fp16`).

R_{px}	W	Direct W^2	`cuFFT` (zero)	`cuFFT` (clamp)	Sep. $R = 4$		Sep. $R = 8$		Sep. $R = 12$
		`fp32`	`fp32`	`fp32`	`fp32`	`fp16`	`fp32`	`fp16`	`fp32` / `fp16`
8	21	2.91	0.92	1.34	1.27	**0.59**	2.60	1.22	3.91 / 1.90
16	41	9.79	1.22	1.22	2.06	**1.16**	4.14	2.35	6.24 / 3.58
32	79	35.73	**1.10**	1.58	3.73	2.01	7.50	4.15	11.28 / 6.22
64	155	136.70	**1.45**	1.87	7.08	3.84	14.16	7.77	21.26 / 11.69

growth) and yields a conserved positive discrete energy on all nonzero modes. Standard 7-point FDTD ($\nu \le 1/\sqrt{3}$) suffers 19.88% axial phase-velocity error by $\|\mathbf{k}\|_2 = 0.80\pi$, motivating dispersion-relation-preserving and high-order room-acoustics FDTD designs [66, 42, 72, 31, 32]. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\mathrm{ax}}(\mathbf{k}) = 2{+}A(k_x){+}A(k_y){+}A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k) - 1)$ has zero dispersion along the coordinate axes, along $(1,1,1)/\sqrt{3}$ it gives $2{+}6(\cos(\nu k/\sqrt{3}) - 1) = 2 - \nu^2 k^2 + \frac{1}{36}\nu^4 k^4 - \ldots \neq 2\cos(\nu k) = 2 - \nu^2 k^2 + \frac{1}{12}\nu^4 k^4 - \ldots$, leaving an irreducible $O(\nu^4 k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\mathrm{ax}}(\pi,\pi,\pi) \ge -2$ forces $A(\pi) \ge -4/3$ ($\nu k_c \le \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k) - 1)$).
- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [42] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \le 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2$, $\frac{1}{8}\nu^2$, $\frac{1}{16}\nu^2$ and center weight $2 - \frac{7}{2}\nu^2$, which is *identically*

$$M_{\mathrm{IWB}}[\mathbf{m}] = (2 - 4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\, 2^{-(m_x^2 + m_y^2 + m_z^2)} \qquad (\mathbf{m} \in \{-1, 0, 1\}^3),$$

  the ($n_{\mathrm{re}} = 1, n_{\mathrm{cx}} = 0, L = 1, R = 1$) single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.38% diagonal phase-velocity error by 0.80π (Figure 4a,d).

Brillouin-octant Chebyshev optimization and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0 \delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2 + m_y^2 + m_z^2)$ (param-

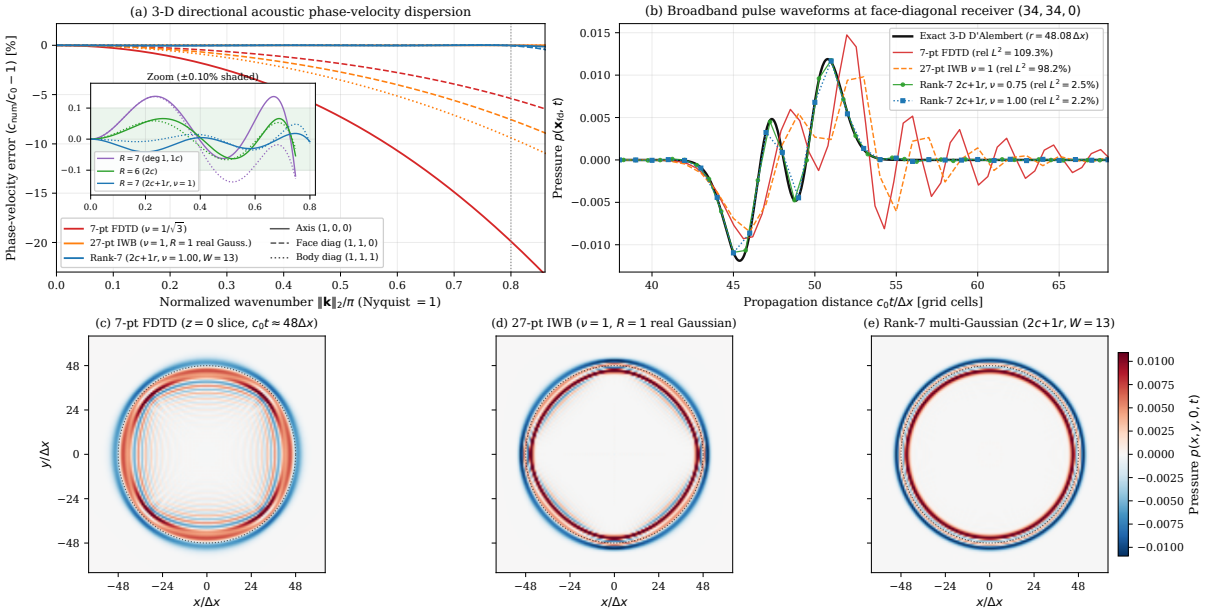


Figure 4: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error ($c_{\text{num}}(\mathbf{k})/c_0 - 1$) along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

eters in Table 10), where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = n_{\text{re}} + 3n_{\text{cx}}$ (Proposition 2.8) or $R = 7$ for $\deg P = 1$ (Proposition 2.9; verified to $\leq 5.6\times10^{-16}$ on $[-L, L]^3$). For fixed rates (searched via differential evolution and Nelder–Mead), minimizing $\max_{0<\|\mathbf{k}\|_2\leq k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)|/(2\nu\|\mathbf{k}\|_2 \sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound speed $\partial^2_{k_x}\widehat{M}(\mathbf{0}) = -2\nu^2$ and strict interior stability $-2+10^{-3} \leq \widehat{M}(\mathbf{k}) \leq 2 - \min(10^{-3}, \frac{1}{4}\nu^2\|\mathbf{k}\|_2^2)$ across 232,620 symmetry-inequivalent Brillouin-octant and spherical-shell wavevectors (including the exact sphere $\|\mathbf{k}\|_2 = k_c$) is solved via cutting-plane HiGHS linear programming, followed by multi-start continuous Nelder–Mead maximization of $|\delta c(\mathbf{k})/c_0|$ over $0 < \|\mathbf{k}\|_2 \leq k_c$. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all $3R$ 1-D passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities (4.5×10^{-7} max `float32` energy drift over 1,000 steps in a 64^3 room).

Table 8 and Figure 4 summarize accuracy and GPU timings (*at $W = 17$, cuDNN switches `conv3d` to an FFT algorithm). Comparing all relevant baselines side by side clarifies both the strengths and the practical limits of high-rank radial stencils:

- **Within separable spatial stencils:** Replacing naive $(n_{\text{re}} + 4n_{\text{cx}}) = 9$ expansion by Proposition 2.8's exact $R = 7$ formula saves 21.0% wall-clock time (5.59 ms vs. 7.08 ms in `float32`, 41.2× faster than direct 13^3 `conv3d`). On the 256^3 broadband spherical pulse test ($\sigma_0 = 1.40\,\Delta x$), 7-pt FDTD and 27-pt IWB break into anisotropic dispersive wave trains (109.3% and 98.2% waveform error), whereas `Rank-7` ($2c + 1r$) keeps $|\delta c/c_0|$ at 0.045% ($\nu = 0.75$) and 0.049% ($\nu = 1.00$, i.e. $\leq 0.05\%$, < 1 cent) up to 0.80π (2.47 points per wavelength at $k_{0.1\%} = 0.81\pi$), achieving 1.6%–2.8% broadband waveform error across all axes and diagonals (3.2% with `float16` intermediates at $\nu = 0.75$). Consistent with the small-stencil regime in Table 6, freeing the $(L + 1)$-tap even 1-D discrete factors from the

Table 8: Numerically optimized 3-D acoustic propagators vs. FDTD and periodic k-space `cuFFT` baselines on 256^3 voxels (RTX 4070 Ti Super; explicit parameters in Table 10). Here $|\delta c/c_0|$ is the worst-case phase-velocity error up to k_c (0.80π for finite-difference baselines, 1.00π for k-space `cuFFT`); $k_{0.1\%}$ and $k_{0.5\%}$ are bisected spherical cutoffs keeping $|\delta c/c_0| \le 0.1\%$ and 0.5% across 733 sampled octant directions; Wave L^2 is the worst-case receiver relative L^2 waveform error at $r \approx 48\,\Delta x$ ($\sigma_0 = 1.40\,\Delta x$, where k-space `cuFFT`'s 0.35% reflects only the pulse's spectral tail outside $[-\pi, \pi]^3$); GPU step times compare direct W^3 `conv3d`, naive $(n_{\mathrm{re}} + 4n_{\mathrm{cx}})/9$-term Triton passes, and each method's primary `fp32` and `fp16`-intermediate step.

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	Step `fp32`	Step `fp16`
7-pt FDTD	0.58	3	—	19.88%	0.060π	0.135π	109.3%	—	—	0.323 ms	—
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.74%	0.099π	0.222π	—	—	—	—	—
27-pt IWB ($1r$)	1.00	3	1 (1)	9.38%	0.085π	0.191π	98.2%	—	—	0.323 ms	—
k-space `cuFFT` (per.)	1.00	256	—	0.000%	1.00π	1.00π	0.35%	—	—	2.73 ms	—
Rank-4 ($1c$+$1r, 0.75\pi$)	0.75	9	4 (5)	0.293%	0.14π	0.76π	11.6%	76.9 ms	4.05 ms	3.30 ms	2.39 ms
Rank-6 ($2c, 0.75\pi$)	0.75	11	6 (8)	0.066%	0.76π	0.80π	5.2%	139.0 ms	6.30 ms	4.80 ms	3.44 ms
Rank-7 (deg 1 $1c, 0.75\pi$)	0.75	13	7 (9)	0.138%	0.16π	0.78π	7.1%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-7 ($2c$+$1r, 0.80\pi$)	0.75	13	7 (9)	0.045%	0.81π	0.85π	2.8%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-7 ($2c$+$1r, 0.80\pi$)	1.00	13	7 (9)	0.049%	0.81π	0.87π	2.5%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-9 ($3c, 0.85\pi$)	0.75	17	9 (12)	0.044%	0.86π	0.89π	2.0%	104.6* ms	10.02 ms	7.66 ms	5.43 ms

continuous Gaussian formula under the same octant LP/Nelder–Mead objective substantially improves the smallest stencil (`Rank-4`, $W = 9, L + 1 = 5$: 0.293% → 0.108%) while yielding diminishing gains as L grows (`Rank-6`, $W = 11, L + 1 = 6$: 0.066% → 0.056%).

- **Comparison with 1-pass compact FDTD and with global `cuFFT` k-space steps:** On a fixed 256^3 grid, single-pass 27-pt IWB runs at 0.323 ms/step (51.9 GVox/s, 92% of peak GDDR6X bandwidth)—12.5× to 17.3× faster per step than `Rank-7` (4.03 ms `fp16` / 5.59 ms `fp32`) and using 3 field buffers rather than 6 in our Triton benchmark (5 if accumulating into p^{n-1} in place). When coarse dispersion (2%–9%) is acceptable, compact 27-pt/25-pt schemes [42, 31] remain faster and simpler; `Rank-7` wins over IWB only when strict wideband phase accuracy is required, where an analytical $(\Delta x)^{-3}(\Delta t)^{-1} \propto k_{\mathrm{tol}}^{-4}$ scaling estimate implies IWB at $\nu = 1$ would need a 9.53× finer grid per axis (865× more voxels, exceeding 16 GiB VRAM, and an estimated 476×–660× longer runtime at 0.1% tolerance, or 20×–28× at 0.5% tolerance). Conversely, on a *homogeneous periodic or rectangular box* where global spectral transforms apply, the periodic 256^3 `torch.fft.rfftn`/`irfftn` k-space step [65] in Table 8 takes just 2.73 ms in `float32` with zero phase-velocity dispersion up to π (0.35% pulse L^2 error)—beating `Rank-7` in both speed and accuracy! Local separable stencils occupy a middle niche only where global all-to-all FFTs are undesirable (e.g. multi-GPU domain decomposition with L-cell halos or locally varying time steps), subject to the boundary and heterogeneity challenges discussed below.

7 Limitations, Uncertainties, and Open Problems

To avoid overstating either the theoretical scope or the practical applicability of our results, we summarize six limitations, trade-offs, and remaining uncertainties:

1. **Continuous-domain exactness, discrete aliasing, and non-certified heuristic optimization:** Theorems 2.3–2.5 classify *exact* representations of *continuous* radial kernels on $\mathbb{R}^d$ (leaving $\varphi(s)$ for $s < 0$ unconstrained). Smooth non-exponential-polynomial targets ($e^{-\|x\|^4}$, Airy patterns $(J_1(r)/r)^2$, ideal circular low-pass filters) have infinite exact rank, while discontinuous kernels require separate arguments ($\mathbf{1}_{\{\|x\|=0\}}$ has rank 1; $\mathbf{1}_{r\le 1}$ has infinite rank). Sampling a continuous rank-R kernel on $\mathbb{Z}^d$ yields discrete rank *at most* R (strictly lower at aliased frequencies $\operatorname{Im}\lambda_u \in \pi\mathbb{Z}$ or when $W < R$). Moreover, while our

inner Chebyshev weight solver is an exact linear program, outer optimization over nonlinear rates (α_u, ω_u) is nonconvex and heuristic: for 6c disc blur ($R = 12$), `blur_2d.py` reaches ripple ± 0.00193 whereas dedicated Remez exchange in [58] reaches ± 0.0014, and in 3-D acoustics our stability check ($|\widehat{M}(\mathbf{k})| < 2$ across 232,620 Brillouin-octant/shell points plus multi-start continuous Nelder–Mead searches) is a dense numerical audit rather than a formal interval-arithmetic or sum-of-squares proof over all $\mathbf{k} \in [-\pi, \pi]^3 \setminus \{\mathbf{0}\}$.

2. **Visual and rendering limitations in 2-D lens blur:** Truncating 1-D passes to $[-S, S]$ produces a square $[-S, S]^2$ support ($r \le \sqrt{2}\, S$), and flat-top approximations built from oscillatory complex Gaussians inevitably exhibit small negative stopband ripples ($\pm\delta$). Around extreme HDR impulses ($10^3\times$–$10^5\times$ scene white), negative lobes can create dark ringing halos unless clamped or tone-mapped, whereas positive Monte Carlo disc gathering is strictly non-negative. Separable convolution is also inherently shift-invariant: real depth-of-field rendering with spatially varying circle-of-confusion $R_{\text{px}}(x, y)$ suffers foreground/background occlusion bleeding unless paired with depth-layer decomposition [27], and for purely shift-invariant large-radius blurs ($R_{\text{px}} \ge 32$), padded 2-D `cuFFT` (1.10–1.87 ms on 1920×1080 RGBA, Table 7) is faster than direct multi-pass FIR cascades.

3. **Conditioning of root-of-unity identities at large d or high multiplicity k:** In Propositions 2.8 and 2.9, summing M alternating products with prefactor $\frac{2^{d-1}}{M}$ bounds worst-case term amplification for unit-modulus factors by 2^{d-1}: benign for $d = 2, 3, 4$ ($2^{d-1} = 2, 4, 8$), but exponential in $d \gg 1$. For degree $k = \deg P \ge 1$ on $[-S, S]^d$, evaluating $U_j(X)$ on $|X| = 1$ mixes $O(1)$ and $O(S^{2k}/k!)$ coefficients; rescaling $X \mapsto \rho X$ with $\rho \approx S^{-2}$ ($\rho^{-k}[X^k] \prod_j U_j(\rho X)$) balances monomial magnitudes, while balanced roots ($|\beta|^{1/d}$) equalize mode norms. Near coalescing rates (compare the $O(h^{-1})$ weights in Section 3.4), $d = 2$ permits exact *a posteriori* $\mathrm{GL}_R(\mathbb{R})$ orthogonalization across components on $\mathrm{rowSpan}(K)$ (Figure 2c), whereas the lack of transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations in $d \ge 3$ favors well-separated complex Gaussians ($k = 0$) over high-degree repeated rates.

4. **Wide-stencil obstacles, boundaries, and global FFTs in 3-D acoustics:** While `DCT-II` half-sample reflection handles homogeneous rectangular rigid rooms cleanly, wide stencils ($L = 4$–8 grid cells in `Rank-4`–`Rank-9`) reach across thin walls or baffles and cannot be applied directly at staircased, curved, or frequency-dependent impedance boundaries without one-sided boundary closures whose passivity and stability for signed multi-Gaussian stencils are currently unknown. On simple homogeneous boxes where no interior obstacles exist, global `cuFFT` k-space propagation (2.73 ms/step, Table 8) is both faster and dispersion-free; extending separable radial stencils to heterogeneous sound speed $c(\mathbf{x})$ and stable boundary closures is necessary before they can compete in realistic room geometries.

5. **Scope of formal verification and cross-literature connections:** Lean 4 and Isabelle/HOL guarantee that the $22 + 1$ formal theorems follow logically from their definitions and standard classical axioms, but human inspection remains essential to verify that those definitions (`IsSumOfProducts`, `separableRank`, `IsRepresentation`) faithfully capture the intended mathematical objects, and none of the steerable pass-count arguments, Python/Triton GPU code, or numerical tables in Section 6 are formally verified. Likewise, line-count comparisons reflect library state and author fluency rather than intrinsic proof length, and while (9) links our exact rank formulas directly to Kronecker ($d = 2$), Winograd–Alder–Strassen ($d = 3$), and Bläser–Mayer–Shringi ($d \ge 4$) algebra-complexity lower bounds [74, 1, 10], special cases of the explicit root-of-unity functional syntheses (Propositions 2.8–2.9) may also appear under different terminology in other communities.

6. **Open problem — sharpness of the general compact-open border-rank bound in $d \ge 3$:** Theorem 2.7 proves that every representation $\mathcal{R}$ has compact-open real border rank at most $B_d(\mathcal{R}) = C_d(\mathcal{R}) - (d-2)\sum_u \deg P_u$ (and complex border rank at most

$\sum_{\mu\in S}(\deg P_\mu+1)=\dim_{\mathbb{C}}\mathcal{A}_\psi$). For a single real degree-1 component $(0,X)$ ($K(x)=\|x\|^2$), $B_d=2$ is sharp (W-state border rank 2 [8, 14]). Whether $B_d(\mathcal{R})$ equals the exact compact-open real border rank for all nonreal and higher-degree radial profiles in $d\geq 3$—for instance, whether $\|x\|^2\cos(\|x\|^2)$ can be approximated uniformly on compact sets below rank $2d$ (6 in $d=3$) by non-Gaussian separable sequences (compare border-rank and border-subrank results on local algebras [78, 16])—remains open.

A Explicit Filter and Propagator Coefficients

To allow standalone reproduction of the main designs in Section 6 from this PDF (complete JSON archives for all 18 blur variants and all 6 acoustic propagators are provided in `Applications/`), every radial profile in Tables 9–10 is tabulated in the single canonical convention of Definition 2.2:

$$\begin{aligned}\varphi(s)&=\sum_{u=1}^{m}\mathrm{Re}\left(\left(\sum_{q=0}^{k_u}(p_{u,q}+iq_{u,q})s^q\right)e^{(-\alpha_u+i\omega_u)s}\right)\\&=\sum_{u=1}^{m}\sum_{q=0}^{k_u}s^q\big(p_{u,q}\cos(\omega_u s)-q_{u,q}\sin(\omega_u s)\big)e^{-\alpha_u s},\end{aligned}\tag{16}$$

where real-rate components satisfy $\omega_u=0$ and $q_{u,q}=0$. In 2-D circular disc blur (Table 9), $s=(x^2+y^2)/R_{\mathrm{px}}^2$ on $|x|,|y|\leq(1+\Delta r)R_{\mathrm{px}}$ ($\Delta r=0.2$), normalized in 2-D discrete execution so $\sum_{x,y}K[x,y]=1$. In 3-D acoustic leapfrog propagation (Table 10), $M[\mathbf{m}]=\alpha_0\delta_{\mathbf{0}}[\mathbf{m}]+\varphi(m_x^2+m_y^2+m_z^2)$ on integer grid offsets $\mathbf{m}\in[-L,L]^3$ ($W=2L+1$), where $\alpha_0:=2-\sum_{\mathbf{m}\in[-L,L]^3}\varphi(\|\mathbf{m}\|^2)$ must be computed from the evaluated φ grid so $\widehat{M}(\mathbf{0})=2$ holds to machine precision (avoiding exponential constant-mode blow-up $\widehat{M}(\mathbf{0})>2$ and preserving spatially constant initial states; note $\zeta=1$ is a double root of $\partial_{tt}\bar{p}=0$, so nonzero initial velocity mean gives physical $a+bn$ motion). Table 10 lists the exact α_0 recomputed from the 9-decimal coefficients (within 2.2×10^{-8} of `float64` JSON), matching $\partial_{k_x}^2\widehat{M}(\mathbf{0})=-2\nu^2$ to $\leq 8\times10^{-8}$ and keeping $\widehat{M}(\mathbf{k})\in[-1.9990,1.9946]\subset(-2,2)$ on the verification grid.

Code and Data Availability

All source code, machine-checked proofs, numerical optimization scripts, GPU benchmarks, JSON parameter/result archives, and figure-generation scripts for this work are openly available in the public domain (CC0 1.0 Universal) at `https://github.com/OlliNiemitalo/kernel-rank-lean`:

- `KernelRankLean/` and `Experiments/`: Lean 4 (`v4.36.0-rc2` with Mathlib `v4.36.0-rc2`) formalizations of all d-D and 2-D separable-rank, root-of-unity, differential-annihilator, border-rank, and alternative semigroup theorems;
- `Isabelle/`: Isabelle2026 (`HOL-Nonstandard_Analysis`) formalization `Semigroup_NSA.thy`;
- `Applications/`: reproducible Python, SciPy, PyTorch, and Triton scripts (`blur_2d.py`, `acoustics_3d.py`, `generate_figures.py`) and saved JSON result archives (`results_blur_2d.json`, `results_acoustics_3d.json`);
- `scripts/`: hermetic setup (`scripts/setup`), formal verification and axiom/oracle audit (`scripts/check`), isolated GPU Python wrapper (`scripts/python`), and PDF build (`scripts/build-paper`).

Table 9: Explicit canonical coefficients (16) for representative 2-D circular disc-blur designs ($\Delta r = 0.2$, Table 5).

Design ($R = C_2$, C_3, $\pm\delta$)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
$1c$ (R=2, C_3=3, $\pm$0.2324)	1 (0)	0.862715	1.624792	+0.767553	−1.862917
$2c$ (R=4, C_3=6, $\pm$0.0759)	1 (0)	0.894640	5.265923	+0.413699	+0.555222
	2 (0)	1.973219	1.554245	+0.510373	−4.611414
$1c_2$ (R=4, C_3=7, $\pm$0.1165)	1 (0)	2.878934	3.152485	+0.883521	−7.976371
	1 (1)	2.878934	3.152485	−16.607444	+7.931853
$2c$+$1r$ (R=5, C_3=7, $\pm$0.0446)	1 (0)	0.962816	7.154625	+0.091824	−0.456790
	2 (0)	2.075279	3.285822	−3.434558	+0.588375
	3 (0)	2.494577	0	+4.387353	0
$4c$ (R=8, C_3=12, $\pm$0.0109)	1 (0)	4.338835	1.553577	−5.754013	−46.161778
	2 (0)	3.838091	4.692138	+9.776587	+15.218676
	3 (0)	2.790788	8.181910	−3.042741	−0.293993
	4 (0)	1.345593	12.330086	+0.009312	−0.245279
$6c$ (R=12, C_3=18, $\pm$0.0019)	1 (0)	5.143786	2.080541	−82.448179	−111.252655
	2 (0)	5.613157	6.153062	+113.961587	−58.149957
	3 (0)	5.982623	9.803250	+39.766911	+161.948149
	4 (0)	6.506533	11.060675	−71.545624	−94.857314
	5 (0)	3.871248	14.810010	+1.416029	+3.711177
	6 (0)	2.202364	19.033594	−0.152648	+0.108514

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This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [58] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [54, 55, 56, 57]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, suggested extending the formalization to d dimensions alongside the visual pipeline exposition and 2-D/3-D GPU applications, and shared independent related-work reviews that sharpened the commutative-algebra and operator-theory connections. We also thank the aiXiv Official Agent reviewers on versions 1.0, 1.1, and 1.2, whose critiques motivated Theorems 2.5 and 2.7, Proposition 2.9, the notation harmonization, the standalone Limitations section, and Appendix A. Both the repository and this manuscript are released under CC0 1.0 Universal.

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Table 10: Explicit 9-decimal canonical coefficients (16) for all six 3-D acoustic leapfrog propagators in Table 8 (set $\alpha_0 := 2 - \sum_{\mathbf{m}\in[-L,L]^3} \varphi(\|\mathbf{m}\|^2)$; reference α_0 shown for verification).

Scheme (ν, W, R, α_0)	u (q)	α_u	ω_u	$p_{u,q}$	$q_{u,q}$
Rank-4 (1c+1r, ν=0.75, W=9, R=4, α_0= − 1.440427676)	1 (0)	0.439979390	0.502750129	+0.137876459	+0.024929224
	2 (0)	0.888101329	0	+0.492695925	0
Rank-6 (2c, ν=0.75, W=11, R=6, α_0= − 1.419560209)	1 (0)	0.606539114	0.240982835	+0.597581203	+0.180225818
	2 (0)	0.267859019	0.385016560	−0.007190817	+0.050272529
Rank-7 (deg 1 1c, ν=0.75, W=13, R=7, α_0= − 1.434649594)	1 (0)	0.429753880	0.229296784	+0.605530664	+0.525638174
	1 (1)	0.429753880	0.229296784	−0.061586407	−0.061209821
Rank-7 (2c+1r, ν=0.75, W=13, R=7, α_0= − 1.439243312)	1 (0)	0.381717837	0.311635272	+0.110431944	+0.263079509
	2 (0)	0.163094730	0.300649461	−0.015761751	−0.008519990
	3 (0)	0.599866965	0	+0.491386225	0
Rank-7 (2c+1r, ν=1.00, W=13, R=7, α_0= − 1.959018821)	1 (0)	0.321809087	0.262377242	+0.159170237	+0.129231489
	2 (0)	0.126172334	0.278598443	−0.010612383	+0.004174471
	3 (0)	0.438442015	0	+0.241253606	0
Rank-9 (3c, ν=0.75, W=17, R=9, α_0= − 1.447076570)	1 (0)	0.301382846	0.246772855	−0.286641369	+0.234406754
	2 (0)	0.398975160	0.094363260	+0.866865360	+0.865008109
	3 (0)	0.111967878	0.275076161	−0.002296910	−0.011770692

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